



BANGLADESH UNIVERSITY OF PROFESSIONALS (BUP)

ResQ: A Disaster Management System

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5. Date of Submission:	July 18, 2025
6. Submitted To:	Lec Aditya Rajbongshi, Lec Rubel Sheikh
7. Course Title:	Software Development Project – I
8. Course Code:	CSE-3208

Abstract

The "ResQ" system is an innovative and innovative disaster relief and management solution that reacts to disasters by linking victims and volunteers and giving them quick relief. The system depends on location-based services to deliver quick assistance, track the delivery of relief, and provide important information in a time of crisis. It supports both volunteers and donors, who are able to become actively involved in disaster relief work. Through database management of victims, donations, and volunteers in one place, ResQ enables efficient coordination during emergencies. The database also serves as an essential tool for NGOs and authorities to monitor and provide disaster management services in real time. This paper expounds on the background, purpose, and significance of the ResQ system with focus on its potential to provide life-saving relief as well as create a sustainable partnership response system for future disaster.

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1. Introduction

1.1. Background

Natural and artificial calamities can happen at any time and impact populations, leaving them vulnerable and in need of immediate relief. Coordination between stakeholders—government agencies, NGOs, volunteers, and citizens—is needed during hopeless moments to minimize destruction and loss of life. Traditional disaster relief frameworks tend to be uncoordinated, strained in resource deployment, and inefficient delivery of assistance at the right time. The ResQ system addresses all these issues through a centralized, simple interface that facilitates connection between victims and local volunteers as well as relief agencies in real-time.

The system has some fundamental features such as location-based assistance, a structured database of volunteers, immediate disaster relief services, and real-time tracking of relief efforts, all aimed at the fast delivery of relief to affected individuals. ResQ, by leveraging technology, offers a new way of dealing with disaster and relief.

1.2. Purpose of the project

ResQ project seeks to develop a comprehensive disaster management platform that enables victims to call for help immediately, facilitates volunteers to volunteer easily, and guarantees efficient distribution of relief resources. By consolidating a myriad of disaster response mechanisms into a single platform, ResQ seeks to render disaster relief activities more coordinated and effective. The site also aims to create an avenue where individuals can directly take part in disaster response efforts, either by volunteering or making donations, thereby creating a sense of collective responsibility during times of disaster.

1.3. Scope of the project

The scope of the ResQ project is from both the preliminary and prolonged relief stages of a crisis. The system can assist individuals who are in need because it provides access to local shelters, clinics, and emergency services. Individuals can register as volunteers to provide ground support, while resources may be contributed. It logs donations, monitors relief distribution, and registers

vital information on special food and medical requirements of the victims. The project has a "What to Do" feature that provides vital guidelines for the handling of various types of disasters. The ResQ system is made available to government bodies, NGOs, and the general public and serves as a convergence point for all disaster management activities.

1.4. Objectives

- To provide a location-based platform for victims to receive immediate help during disasters.
- To register and organize volunteers who can assist in disaster relief operations.
- To enable donations and track the distribution of relief materials and financial contributions.
- To provide accessible disaster guidance for common emergency situations.
- To serve as a centralized system that can be accessed by government bodies and NGOs for disaster management coordination.
- To ensure transparency in relief distribution and ensure that resources reach those in need in a timely manner.

1.5. Significance

The significance of the ResQ project lies in its coordination of relief providers with the victims during the disaster period. Offering a platform where both short- and long-term requirements are met, ResQ makes no person go without assistance during the time of disaster. Secondly, transparency offered through tracking donations and relief increases trust in the system. Centralization of disaster management services within the project is particularly beneficial for governments and NGOs, who typically face logistical challenges in emergencies. Providing an integrated system with volunteers, victims, relief resources, and disaster counsel, ResQ facilitates more efficient and collaborative disaster response systems, ultimately saving lives and minimizing the impacts of disasters.

2. Methodology

Technique of development along with execution of 'ResQ' disaster response as well as an administration system, was developed with an aim of dealing with the complexity of disaster response. In view of the complexity of issues, it involved meticulous planning, incorporation of the latest available technology, along with regular testing. It is systematic in nature, with an option of rapid development of functionalities, with a long-term perspective of providing immediate care along with efficient completion of disaster relief operations.

2.1. Planning of the project

Planning of 'ResQ' involved systematic planning along with emphasis on grasping user needs, incorporation of inputs from those provided by stakeholders, as well as scaling of this system in large-scale disasters.

2.1.1. Understand the problem

The key point of planning involves learning disaster disaster-affected community's needs. We interviewed emergency response officers, NGOs, govt agencies, as well as planned end-users (volunteers as well as victims). These helped in designing the system such that it would address issues seen in real life.

2.1.2. Identifying Key Features

Victim and Volunteer Registration: Two-database structure (a victims database and a volunteers database) enables end-users to be properly categorized and matched accordingly based on need, skills, as well as geography. Volunteer mobilization is consequently rationalized as well as streamlined.

Need Help Feature: It is a panic feature allowing members immediate aid. Victims only need to press a panic button on the platform when they need help, summoning rescuers and nearby volunteers.

What to Do Section: In this educational feature, there is advice concerning things one can accomplish under varied kinds of disasters, enabling one to act promptly as well as efficiently in difficult times.

Location-Based Direction: Real-time maps direct victims towards accessible shelter centers, hospitals, firefighting trucks, as well as police command centers in nearby points. It reduces the time spent in finding required resources in disorientation in a disaster.

Volunteer Engagement and Donation Tracking: Through the system, members can easily volunteer, donate, or offer any resources. All the donated resources will be tracked, and the members will know how their contributions have been utilized. Central Coordination Hub: 'Resq' is one of the central coordination hubs of NGOs, along with those of the government. agencies. It enables quick allocation of resources along with quick decision-making.

2.1.3. Iterative Development and Prototyping:

This development process was step-wise, using agile principles in order to deploy as well as build features in small increments. Development began with low-fidelity prototyping, such that ideas would be cycled quickly in order to elicit comments. Iteratively, the system was improved as well as matured such that it accurately represented the scenario of disaster response in near-real-time.

2.1.4. Stakeholder Engagement and Testing

Because the 'Resq' project was first and foremost a college project, there was no in-the-field deployment or interaction with emergency response units or authorities. There was, however, full in-house testing and iterative commentary as part of the developmental process. There was intimate working with guides, professors, and fellow students in order to hone in on features of the final product. There was identification of minute points of refinement as well as confirmation of desirable functionality of the platform with this set of stakeholders.

Testing was conducted development environment with a simulated disaster situation in mock emergency cases. Testing was in verification of whether central operations, such as registration of

users, location-based aid, as well as contribution tracking, functioned well. Testing with mock data as well as mock users allowed testing of systems in response in real-time in some manner, although it did not entail any actual emergency cases.

2.1.5. Final Deployment and Scaling

System deployment was done in the academic domain, mostly as a demo. It was presented as a working prototype with core disaster relief, along with management central functionalities. There was no deployment in a world live scenario, as there was no interaction with live systems or outside agencies such as NGOs or governing bodies.

The scaling of architecture was accounted for in development as one way towards scaling traffic as well as data-flow, in case it was being used as a live disaster response. As it was with this project, never being operational outside of academic environments, precautions were thus only taken in order to have the platform operate under hypothetical high-demand cases, such as under a presumed-disaster scenario. Scaling of architecture did take place, but under hypothetical use as opposed to live-testing under a live emergency situation.

2.2. Technology Stack

The 'ResQ' project was carefully developed using a judicious selection of both front-end as well as back-end technologies. These technologies had been handpicked with utmost care in order to develop a highly functional as well as accessible disaster management system that would efficiently cater to several needs. Materials as well as tools considered in this project had only one goal in mind, such as enabling quick development as well as easy deployment, especially in an academic setting, wherein such systems can be of immense help.

2.2.1. Frontend Technologies

HTML, CSS, and JavaScript: The underlying technologies behind building the front-end of the 'ResQ' platform comprised three primary elements: HTML, CSS, and JavaScript. HTML was critically involved in arranging as well as organizing content presented on web pages with a clear

and definitive structure exhibited throughout the information. In addition, CSS was utilized in order to increase the presentational value of elements with effective styling as well as arrangement in place, resulting in a presentable as well as desirable user experience. In addition, JavaScript was used in order to add some degree of interactivity to web pages such they remained not only responsive in response but interesting as well as dynamic with regards to how they reacted once in use.

Bootstrap: Bootstrap, known as one of the most eminent front-end web development frameworks, was used to ensure that the website would be friendly on mobile devices as well as responsive. It served as one of the most important tools in speeding up the development process in view of its rich collection of already designed elements, such as grids, nav bars, as well as buttons. By using them, it ensured that the platform would have a presentable look regardless of the diversity of devices as well as screen sizes used.

2.2.2. Backend Technologies

PHP: PHP was chosen as the programming language of the back-end of the 'ResQ' system mainly because of its powerful capabilities and prevalence. This allowed for easy and efficient processing of user requests, handling of communication with the database, as well as creating dynamic content necessary for a responsive application. In addition, PHP was instrumental in handling server-side logic such that processing of form submissions was done effortlessly, as well as effortlessly sending and retrieving data to and from the database.

MySQL: MySQL was used as one of the main database systems, specifically intended for storing purposes of user data, such as details of victims as well as volunteers, disaster-based data, as well as any other relevant data that can be asked for. MySQL database systems have a lot of powerful data handling functionalities necessary in dealing with large and complicated datasets, and its excellent connectivity with PHP, one of the most prevalent scripting languages, rendered it a suitable as well as favorable as one of the requirement needs of this project.

2.2.3. Developmental Tools

Visual Studio Code: Visual Studio Code, or VS Code as it is commonly abbreviated, was used as the primary code editor of this project. It is not only designed in a light but extremely efficient way, but also with powerful extensions and complete debugging tools, thereby contributing immensely towards accelerating and improving the whole development process.

Git and GitHub: Git was used as an advanced tool of version control, such that every minor modification of the project was systematically tracked and could be reverted with ease in case it had to be done. In addition, GitHub served as the hub repository in which the code of the project was deposited, thus enabling member collaboration. This arrangement not only permitted easy tracking of project advancements but also played a huge role in easy regulation of version control throughout project development.

2.2.4. Server Environment

XAMPP (Apache Server): The development and testing of ResQ: Chaos to Control were done on a local server environment by using the XAMPP open-source cross-platform web server solution stack package. XAMPP hosts Apache HTTP Server, MySQL, PHP, and Perl together to provide an easy but lightweight environment for testing and web application development.

Apache Server:

Apache HTTP Server served as the web server module to process HTTP requests and send the web pages of the application to end-users during development. It provided a production environment on the local machine to enable it to test and debug server scripts written in PHP efficiently.

Local Deployment:

The configuration was deployed locally within the XAMPP environment for facilitating iterative development and ready testing of new functionality in the absence of internet access or remote server setup. Project files were served within the htdocs directory of Apache, allowing developers to access the application through localhost URLs.

Testing and Configuration:

XAMPP simplified the process of server configurations such as virtual host configuration, MySQL database handling, and PHP configuration in the php.ini file. This was especially useful in

debugging problems related to the integration of the backend, connectivity to the database, and form validation before deployment onto a potential production server.

Use of XAMPP as a server platform provided the developers with a productive and flexible environment to test and develop the web application so that all functions were rigorously tested in an environment under their control.

2.2.5. Payment Integration

Dummy Payment System: An area devoted to payment was initially designed as a part of the complete system design; however, it is worth stating that at present, it is not working as designed. This payment section was added in the system as a placeholder, with plans in place such that it would later permit features, such that in the future, it would be possible to facilitate the processing of donations. As it stands at this specific point of development, it literally is being used as a mock feature, as opposed to being a complete working element of the system.

2.3. Project timeline

The project was completed in a successful manner under the allotted academic period, showing strict adherence to a clearly spelled out and accurately detailed timetable of development. Adherence to this schedule allowed for a systematic and orderly process of creating the 'Resq' platform in such a way that every step fell into place efficiently. The exact period of this project is detailed as follows in elaboration:

Phase 1 - Conceptualization and Planning (2 Weeks):

At this first stage of the project, it was a significant step in clearly establishing the project scope, as well as determining essential features necessary for its successful completion. Brainstorming was done as a team effort in order to discuss and thoroughly examine the needs of the platform, and consequently, they produced detailed wireframes in order to clearly define the design layout of the platform with due consideration of every element.

The initial functionalities covered in this discussion encompass core features such as registration of users, handling of victims, as well as volunteers, together with core functionalities essential in successful disaster response operations.

Phase 2 - Design and Prototyping (2 Weeks):

This is one of the phases in the detailed process of making mockups, along with prototypes, showing how the user interface will look. In order to design the general structure of the website, there was an effective use of HTML, CSS, as well as JavaScript.

Feedback was gathered and aggregated from classmates as well as professors in order to refine and enhance the interface effectively, as well as facilitate a better user experience, simply known as UX.

Phase 3 - Development (4 Weeks):

Front-end development was done through an integration of technologies, namely HTML, CSS, JavaScript, and Bootstrap, which have been used aptly to develop extremely responsive and presentably sound web pages.

PHP was used in the back end in order to efficiently handle and run data processing operations and interact with the MySQL database in order to handle data in an efficient manner. Key functionalities that serve as a backbone of the system, including those of victim registration, volunteer registration, and location-based assistance facilities, have been implemented in a clever manner and assimilated into the complete scheme of things of the system.

The database was created using MySQL, a powerful and commonly used relational database management program, in a way that would effectively handle user data. It was done specifically as a means of properly managing information related to victims as well as volunteers.

Phase 4 - Testing and Refinement (2 Weeks):

After initial development levels, the platform underwent a detailed testing process using mock data in order to ensure that every single feature was working as designed. During this essential period, the system was thoroughly tested in order to locate any possible bugs, and any problems encountered as a result of these testing procedures were corrected in due course in order for it to work in an optimum state.

Feedback from peers as well as other team members was gathered in order to refine the user interface considerably, which would create an easier and smoother interaction experience with a varied set of devices.

Phase 5 - Finalization and Deployment (2 Weeks):

The final copy of the project was thoroughly set up before submission in order to satisfy all mandatory standards. This is an involved process of making sure everything worked as required and setting up a complete working demo in order to present its functionality.

The project was thoroughly and efficiently executed in the immediate academic setting, in which its effect was notably demonstrated in front of a varied population of teachers as well as fellow students who had gathered in order to witness.

Phase 6 - Future Improvements (Ongoing):

Although originally completed as a college assignment, there had been lengthy debates on potential upgrades as well as further features available in potential upgrades in the future. In such potential ways of expanding, some interesting points of discussion have been payment systems integrations, as well as integrating data capturing capabilities in a real-time manner.

The Gantt chart of Fig. 2.1 illustrates the ResQ Project timeline.

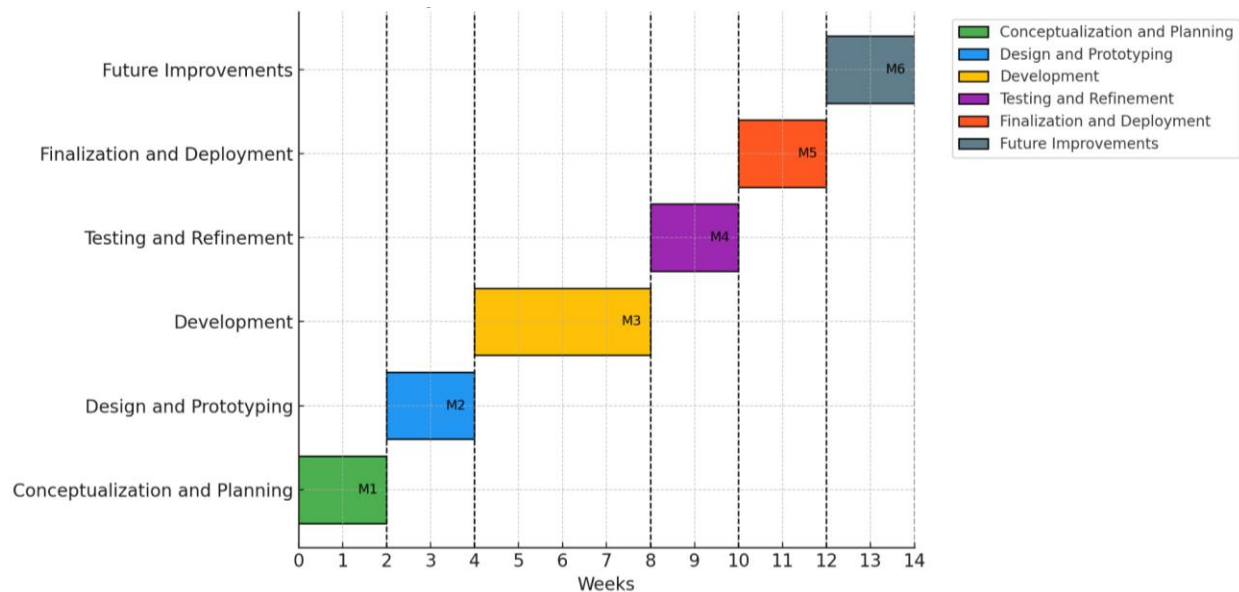


Fig. 2.1. Project Timeline

2.4. Data collection and analysis methods

Initially, the 'Resq' system was developed as a research-based prototype specifically designed for research purposes, along with several research goals. It is necessary to state clearly that throughout none of its operational phases was there any direct data or analytics of actual world figures recorded. However, it was designed in such a way as to facilitate the gathering of essential data inputs originating from users, while at the same time processing such data in ways completely in agreement with and serving the larger goals related to disaster management as well as response efforts.

2.4.1. Data Collection

- **User Registration Data:**

At any point in the process of registration, individuals, including those affected, categorized as victims, as well as those coming forward in order to help, categorized as volunteers, have to present a series of personal details. These details usually include vital information such as full name, relevant contacts, their geographical position, as well as, in this case, especially for victims, any special needs or medical needs they might have. All this data gathered from individuals is kept in a MySQL database in complete security, playing a crucial role: it helps in matching victims with qualified volunteers as well as qualified shelters capable of accommodating them with their specific needs.

- **Victim and Volunteer Data:**

Volunteers contribute in an active manner by posting detailed information about their specific skills as well as dates they can contribute, while simultaneously victims in need present necessary details of the nature of their emergency. The system reviews this valuable information in order to efficiently extend relevant aid or facilitate persons in need with volunteer assistance, with due consideration of geographical location as well as need levels involved with each specific situation.

- **Requests for Help:**

Those who happen to find themselves in dire circumstances have the choice of putting in a "Need Help" request, formally entered and documented within the system. It is extremely detailed in recording such things as the nature of the request being submitted, along with the exact location from whence assistance is being requested, as well as any particular needs that a person may have,

such as medical attention or temporary accommodation of a protected sort. All of this gathered information can potentially be analyzed in depth in order to refine and fine-tune response times in similar occurrences in the future; it is worth mentioning, though, that in this case, no actual-world analytics were actually run or in place.

2.4.2. Data Analysis

Although in its current incarnation the system does not yet enjoy sophisticated functionalities in view of data analytics, it is worth mentioning that the data thus far accumulated can in fact undergo detailed analytical treatment such that there can be an in-depth exploration of several factors at once. In particular, this entails being able to study how efficient volunteer efforts can be in one setting or another, being able to ascertain how fast aid is served in emergency situations, as well as examining how resources have been distributed in a disaster scenario as it is played out in real life. Applied in the context of a working operational framework, the following kinds of data would be analyzed with utmost care and parsed in accordance with their related domains:

- **Response Time Analysis:** In this section, we analyze how quickly volunteers respond to calls for help issued by victims.
- **Volunteer and Victim Utilization:** It is defined as the number of volunteers along with victims who are utilizing the platform in an active way at any point in the disaster.
- **Resource Allocation:** Tracking where relief materials or assistance are allocated based on need. Since initially the system was designed primarily as a demonstration project, it is worth observing in this context that data analysis had not been actively integrated at this design stage or into its functionality. Nonetheless, the concept of adding powerful data analysis features was identified as an important potential future direction of this work, particularly in the case of scaling up this system towards deployment in real-world applications.

3. System Design

3.1. Architecture

ResQ architecture has a three-tier web application architecture: Presentation Layer, Application Layer, and Data Layer. Layered architecture enables the separation of concerns, improved maintainability, and future system scalability. Modularity, scalability, and user accessibility are emphasized in design to ensure effective disaster response and relief coordination.

3.1.1. Presentation Layer

The presentation layer is responsible for providing an interactive and user-friendly interface to the users. It is implemented in HTML, CSS, Bootstrap, and JavaScript to provide a responsive layout on various devices, including desktops, tablets, and mobile phones. Bootstrap makes it responsive on mobile phones, and JavaScript is used to deliver dynamic content generation and interactivity for features like real-time updating of disaster data and interactive mapping.

3.1.2. Application Layer

The application layer contains the system's business logic. It is written using PHP, which handles user requests, data crunching, and database interaction. PHP scripts manage functions such as:

- User (Volunteers and Victims) registration and authentication, along with session management.
- Data validation and processing.
- Interaction with the database.
- Generation of dynamic responses for the frontend.
- Tracking and reporting disaster incidents.
- Relief resource distribution and monitoring.
- Volunteer management and coordination.
- Immediate assistance helpline.
- Emergency disaster combat guidelines.

This layer acts as a front-end-to-back-end gateway to allow smooth data transfer and processing of business logic.

3.1.3. Data Layer

The data layer has MySQL as the backend relational database management system. It holds and manages all the data, including user data, disaster reports, resource inventories, and communication logs. The database is structured to ensure good querying and data integrity, with tables optimized to store:

- User credentials and permissions.
- Disaster incident reports.
- Relief requests and resource allocations.
- Emergency contact information and status.
- Shelter locations and updates.
- Information on the people in need and those assisting them.

Such structuring enables it to be possible to offer a clean separation of data management, application logic, and user interaction, resulting in a highly scalable and robust system.

Database normalization principles are correctly followed in order to reduce redundancy and maintain data integrity.

Figure 3.1. illustrates the architecture of the overall ResQ system.

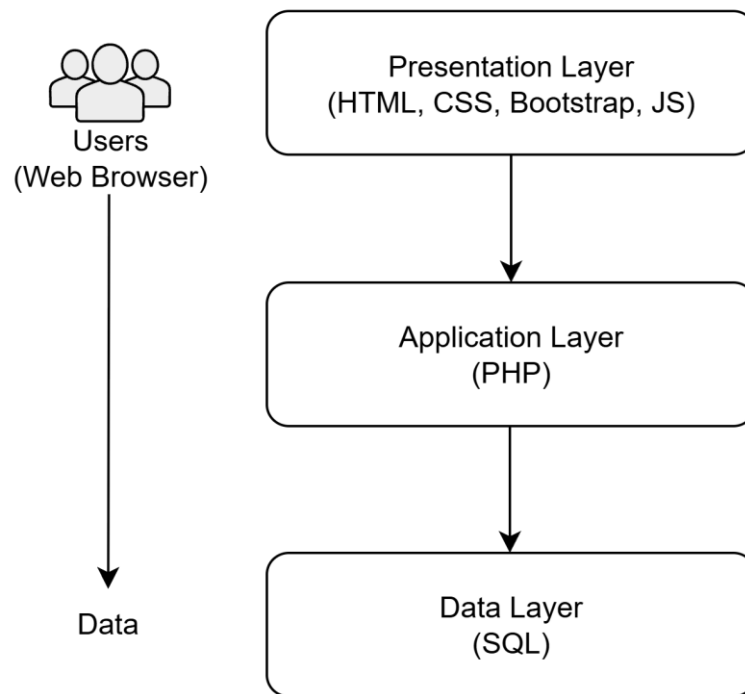


Fig. 3.1. System Architecture

3.2. Design specifications

The system's composition and structure are outlined in detail by the design specification

3.2.1. User Interface Design

The system is comprised of user-friendly interfaces according to different user roles: administrators, coordinators, volunteers, and affected people.

- **System Portal:** Delivers emergency contact information, response guides to different disasters, and immediate helpline search facilities to any user browsing the website.
- **Volunteer Portal:** Offers volunteers the capacity to register, view assignments, and update their availability status.
- **Victim (Affected Persons) Portal:** Enables the reporting of disaster cases and calling for help.

The front end is designed to be responsive to facilitate easy user experiences across multiple platforms and screen sizes.

3.2.2. Database Design

The database schema is organized to support the system's core functions. Major entities and their relationship are described as follows:

Entities and Attributes

1. Disaster (id, name, region, severity, start_date, end_date)
2. Shelter (id, name, location, status)
3. Victim (id, name, address, phone_no, email, occupation, dob, current_location, requirement, medical_info, blood_group, allergy, chronic_disease, password)
4. Volunteer (id, name, address, phone_no, email, occupation, dob, assigned_location, medical_info, blood_group, allergy, chronic_disease, password)
5. Relief (id, type, quantity, location)
6. Police (id, name, contact_no, location, availability_hours)
7. Evacuation Support (id, name, contact_no, location, availability_hours)
8. Medical Response (id, name, contact_no, location, availability_hours)
9. NGO Helplines (id, name, contact_no, location, availability_hours)

Relationships

1. Seeks (Victim → Shelter)

- victim_id → Foreign Key (Victim)
- shelter_id → Foreign Key (Shelter)
- checkout_date

2. Collects (Volunteer → Relief)

- volunteer_id → Foreign Key (Volunteer)
- relief_id → Foreign Key (Relief)
- collection_date
- relief_destination

3. Impacts (Disaster → Victim)

- disaster_id → Foreign Key (Disaster)
- victim_id → Foreign Key (Victim)

4. Involves (Volunteer → Relief → Victim)

- volunteer_id → Foreign Key (Volunteer)
- relief_id → Foreign Key (Relief)
- victim_id → Foreign Key (Victim)

Primary keys and foreign key constraints are used to enforce relationships among tables and referential integrity. Figure 3.2 shows the database schema of the system.

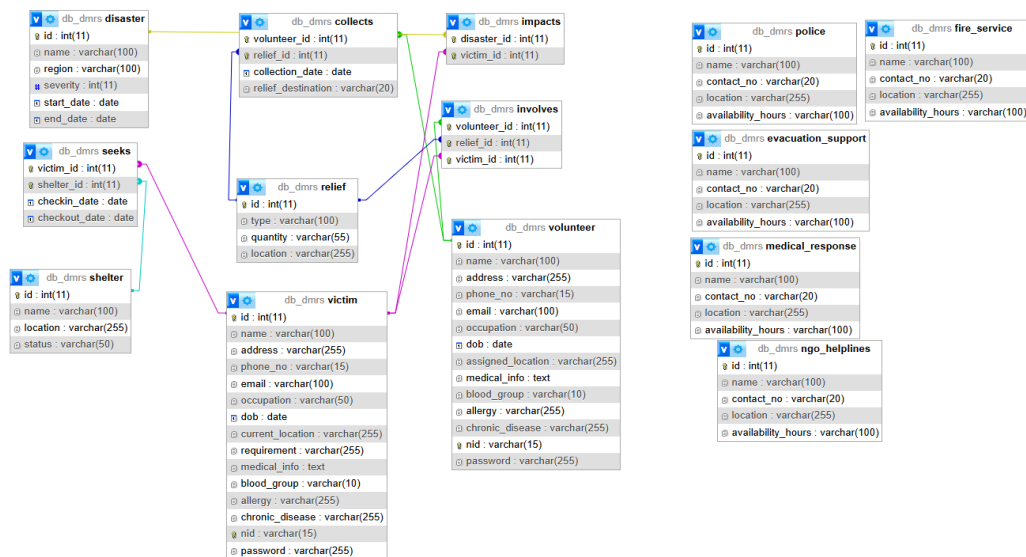


Fig. 3.2. Database Schema

3.2.3. Security Design

Security is enforced at multiple levels:

User Authentication and Authorization: Role-based access control ensures users are given access to functionality only that is relevant to their roles.

Input Validation and Sanitization: Protects against SQL injection and cross-site scripting (XSS) attacks.

Secure Communication: The client-server communication is set to be encrypted in future versions for improved privacy.

3.2.4. Workflow Design

There is a request-response workflow in the system:

1. Users interact with the front end through forms and dashboards.
2. Requests are processed by PHP scripts in the application layer.
3. Data is updated or queried in the MySQL database.
4. Responses are returned to the user interface for display.

Some of the key workflows supported include disaster reporting, resource allocation, and volunteer tracking to facilitate smooth execution during emergencies.

3.3. Development process

The process of development was incremental in nature, in which modules were developed, tested, and refined incrementally.

3.3.1 Front-End Development

The front end has been simplified to use and access for different roles to the users: administrators, volunteers, and disaster victims. Bootstrap has been used to acquire responsive layouts and interactivity for dynamic features like:

- Dynamic updates on disaster occurrences.
- Interactive dashboards for administrators.
- Simple forms for registration and reporting.

3.3.2. Back-End Development

PHP was utilized to create server-side scripts to interact with the requests and control data flow between the front end and the database. Major features included:

- **User Authentication and Role Management:** Secure login and access restriction for different user roles.
- **Disaster Report Handling:** A Facility for users to report disasters with location and severity details.
- **Resource Allocation:** A Facility for administrators to allocate resources to specific disaster zones based on needs.
- **Volunteer Coordination:** Tasking and tracking of volunteer activity.

3.3.3 Database Design

The MySQL database schema was used to facilitate efficient data management. Some of the major features included:

- Relational user, disaster report, resource, and volunteer tables.
- Use of primary keys, foreign keys, and indexes to enhance referential integrity and query optimization.
- Secure storage of data, e.g., hashed passwords and input sanitizing to prevent SQL injection.

3.3.4. Coding and Development Practices

In order to have a proper implementation, the following practices were employed:

- **Separation of Concerns:** Code is split into individual files for presentation, application logic, and database access.
- **Form Validation:** Both client-side (JavaScript) and server-side (PHP) validation are performed to prevent invalid or malicious inputs.
- **Error Handling:** PHP scripts include error handling to log and show user-friendly error messages.
- **Database Optimization:** Indexing and query optimization techniques were employed to increase data retrieval performance.

- **Security Measures:** Input sanitization and prepared statements are employed to avoid SQL injection attacks.

Session management protects user authentication and avoids unauthorized access.

3.3.5. Testing and Debugging

After development, rigorous testing was put in place to achieve system correctness and reliability:

- **Unit Testing:** Front-end modules such as login, registration, and disaster reporting were tested individually.
- **Integration Testing:** Verification of the coordination of front-end form and back-end logic.
- **System Testing:** End-to-end testing was performed in order to see whether the system worked as one unit or not.
- **User Acceptance Testing (UAT):** The system was demonstrated to dummy users, and their input was taken into account in coming up with means of improving the user interface and functionality.

Issues encountered in testing, such as broken links, form validation issues, and database query inefficiencies, were resolved on time to deliver an efficient system.

3.4. Technical challenges and solutions

Throughout the development of ResQ: Chaos to Control, there were a number of technical issues that arose and needed to be carefully analyzed and solved creatively. This section presents the most important problems encountered by the development team and the methods used to successfully overcome them.

3.4.1. Version Control

One of the significant problems encountered was the lack of a good version control system at the beginning of development. As multiple team members were working on different parts of the project simultaneously, there were ongoing problems in maintaining consistency between codebases.

Challenges Faced:

- **Merge Conflicts:** Multiple developers working on identical files (e.g., index.php, styles.css) led to frequent Git merge conflicts.
- **Branch Management:** A Poor branching strategy resulted in mismatches between development and production versions.

Solutions Implemented

To address these issues, the team adopted Git, a distributed version control system. Git enabled the team to:

- Adopted Git Flow Workflow
- Created main (production), develop (staging), and feature branches (feature/login, feature/disaster-reporting).
- Required pull requests (PRs) with peer review before merging.
- Automated Conflict Detection
- Used GitHub's Conflict Editor and git rebase for conflict resolution early on.
- Implement pre-commit hooks to enforce coding conventions.
- Documented Versioning Policy
- Followed Semantic Versioning (SemVer) for releases (e.g., v1.2.0).
- Use commit history to monitor version changes and roll back to earlier states as needed.

Regular use of GitHub as a remote repository also smoothed out collaboration and ensured the safekeeping of project files.

3.4.2. Frontend-Backend Integration

Integration of backend PHP code with frontend components (HTML, CSS, JavaScript, and Bootstrap) was another daunting challenge. Maintaining the data flowing from the user interface to the database in perfect harmony required collaborative effort from both sides.

Challenges Identified:

- Processing asynchronous frontend submissions to PHP scripts.
- Client-side and server-side data validation and sanitizing.
- Debugging frontend form field and backend processing script conflicts.

Solutions Implemented:

The team adopted a modular approach to address this issue:

- Employed AJAX calls where appropriate to provide smooth communication without page reloading.
- Established precise API endpoints in PHP for functionalities such as user registration, disaster reporting, and resource allocation.
- Applied broad debugging methods and browser developer tools to monitor and correct integration bugs within deadlines.

3.4.3. Apache Server Configuration

Installation and setup of the Apache server on the local development environment were also problematic, especially for the new team members.

Challenges Faced:

- Initial problems with the functioning of the Apache server due to conflicting ports.
- Directories became configured in the wrong manner, and PHP was operating inappropriately on the server.

Solutions Implemented:

These issues were resolved by:

- Solving and unblocking port conflicts (most commonly with port 80 or 443) using the use of system commands and XAMPP setting restarts.
- Project files are put in the right position in the Apache htdocs folder for usage.
- Resetting the services in the XAMPP Control Panel and verifying httpd.conf and php.ini settings.
- All these processes helped the Apache server run successfully, thus making it easy to test and deploy the application in the local environment.

3.4.4. Database Design and Normalization

Maintaining a properly designed relational database schema with the capability of handling different entities and their relationships was one of the key challenges.

Challenges Faced:

- Avoiding data redundancy and referential integrity between tables.

- Maintaining complex relationships, such as many-to-many (e.g., multiple volunteers assisting several victims).

Solutions Implemented:

- Applied database normalization constraints to avoid redundancy and maintain the consistency of data.
- Used foreign keys and junction tables (e.g., involves, seeks) for modeling many-to-many relationships in an appropriate manner.
- Carried out extensive testing with test data to validate database queries and constraints.

3.4.5. Responsive UI Design

An important requirement was to make sure the platform is responsive on any device and easy to use.

Challenges Encountered:

- Keeping layouts consistent with varying screen sizes.
- Rendering dynamic content without compromising responsiveness.

Techniques Adopted:

- Employed Bootstrap's grid and responsive utility classes to provide flexible layouts.
- Tested the app on different devices and browsers to identify and fix UI inconsistencies.

3.4.6. Debugging and Error Handling

During development, the system used to act strangely now and then because of faulty logic and wrong configurations.

Challenges Faced:

- The following errors can be traced that were faced at different levels (frontend, backend, and database).
- Processing invalid or incomplete user input that was breaking workflows.

Solutions Implemented:

- Browser development tools and PHP error reporting capabilities were leveraged to monitor and correct problems.

- Standard error-handling routines were used to detect exceptions and provide friendly error messages.

Through the solutions of these technical problems, the development team made sure that ResQ was a stable and friendly disaster management tool. Indeed, by overcoming these hurdles, the team was able to develop an effective and streamlined disaster management system. These solutions not only addressed short-term concerns but also set improved practices in terms of cooperation and system implementation for future initiatives. These experiences also provided the team with improved technical skills and work procedures.

4. Results

4.1. Overview of the Developed System

The ResQ: Chaos to Control web application was developed as a comprehensive disaster relief and management system that would support disaster response activities and help impacted people in an adequate way. The system is implemented in a local development environment now and adequately demonstrates its intended functionalities through a user-friendly and responsive interface.

4.1.1. Key Achievements

End-to-End Functionality: ResQ successfully combines all the modern web technologies to deliver a robust, one-of-a-kind disaster management functionality. The frontend, developed using HTML5, CSS3, Bootstrap 5, and JavaScript, offers interactive forms, real-time dashboards, and location-based services on any device. The backend, based on PHP/MySQL, delivers secure data processing with robust authentication and validation mechanisms for data integrity. This synergy has enabled the creation of an intuitive system, comprehensive dashboards for tracking status, and location-based search interfaces, all optimized to function flawlessly across devices. The solution integrates critical features like a secure authentication module with password hashing, sophisticated query management for location-based services, and comprehensive data validation routines to enable system security and integrity.

Role-Based Access: Another major achievement of the system is its role-based access control, where three distinct kinds of users have personalized experiences. Victims are provided with an uncomplicated interface through which reporting disaster incidents and procuring help becomes simpler, while volunteers are able to access dedicated interfaces for managing tasks and coordinating resources. Unauthenticated members can browse important public data like emergency contacts and preparedness advice, and thus the system is accessible by both registered members and the general public. Advanced access control is facilitated by robust security aspects like secure session management, role-based rendering of UI, secured routes for sensitive actions, and industry-standard password hashing using the bcrypt algorithm.

Local Deployment: Deployment locally using the XAMPP stack has successfully demonstrated the system's viability as a proof of concept. Deployment on Apache HTTP Server, MySQL, and

PHP has confirmed all core system workflows from reporting a disaster through to resolution. The setup has been specifically tailored to assist the development, including optimization of database seeding and configuration files. This local deployment not only proved the technical feasibility of the concept but also illustrated the modularity of the system, which has been created in a manner to enable extension as well as integration of external services if required. The fact that end-to-end system workflows are successfully proven in closed settings offers a solid foundation for further development stages as well as the eventual real-world rollout to prove ResQ to change disaster operations management.

4.1.2. Key Implemented Modules

User Management

The system has individual functionalities for different user roles:

- Victims can log in to the platform to request, for instance, shelter, medical help, and other services.
- Volunteers can log their profiles, set their location, availability, and availability, and be assigned to serve needy victims.
- Guests (non-logged-in users) can get restricted access to certain features, for instance, response protocols to disasters and emergency contacts, to enable necessary information to be accessed by all.

Role-based access control provides safe and effective management of user interactions across the platform.

Volunteer and Victim Coordination

Volunteer and victim communication is done using location, and needs are vital components of the system. Here, the administrators assign volunteers to designated disaster areas and victims to provide timely and effective support.

Resource Allocation

This module allows the management of available resource inventories, such as foodstuffs, medical consumables, and shelter centers. It allows monitoring of available resources, distribution, and replenishment to ensure optimal use of items of aid during response activities.

Shelter Search

Victims may enter their location to access shelters around them and view their real-time availability status. This feature allows affected persons to find safe locations without difficulty in the event of emergencies.

Emergency Contact

Users can input their location to acquire an exhaustive list of emergency contact details regarding their location. This includes fire stations, police stations, medical centers, and evacuation support units to facilitate prompt action in case of an emergency.

Immediate Support Helplines

This module provides real-time access to all helplines available within a locality. With the cataloguing of the facilities, the system enables responders and victims to establish communication links with authorities and organizations in a timely manner.

Disaster Response Guidelines

The website hosts generalized guidelines to disaster response and preparedness, accessible to visitors and registered users. The guidelines provide crucial safety advice and step-by-step process advice on how to minimize the risk of injury in times of disaster.

On the whole, the system developed combines the above modules effectively to offer an all-around disaster management tool. It promotes coordination among victims, volunteers, and administrators, in addition to making key information and resources accessible to the public during crises.

4.2. System Demonstration and Visualization

4.2.1. Home Page

This is the home page of the website, where all the necessary information about the website, what it does, and how to navigate through the website is placed in a systematic and beautiful order.

It contains all the directories needed for a new user to navigate through the website. It has all the crucial links to important pages, a short description of the work done by the ResQ system, which is later broadened in the “About Us” page.



Fig. 4.1. Home Page

4.2.2. User Login & Registration

We have applied a role-based login system for the people who want to help (volunteers) and the people who are in need of help (victims). There are separate login, registration and password recovery methods for the roles.

Role-based Login:

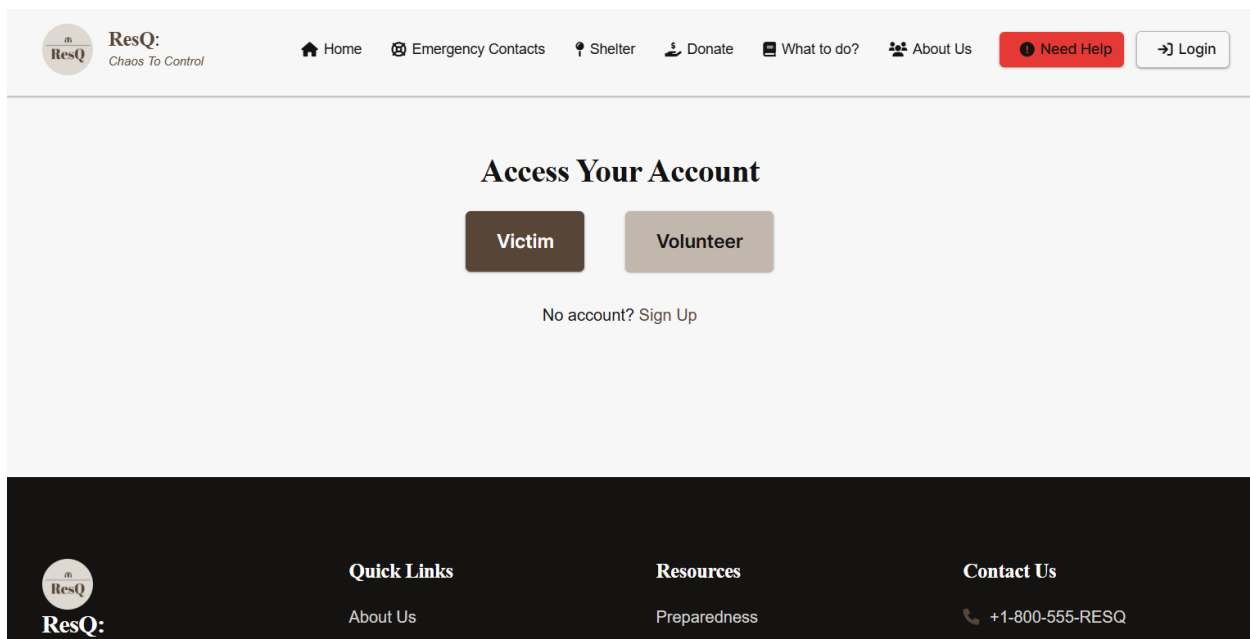
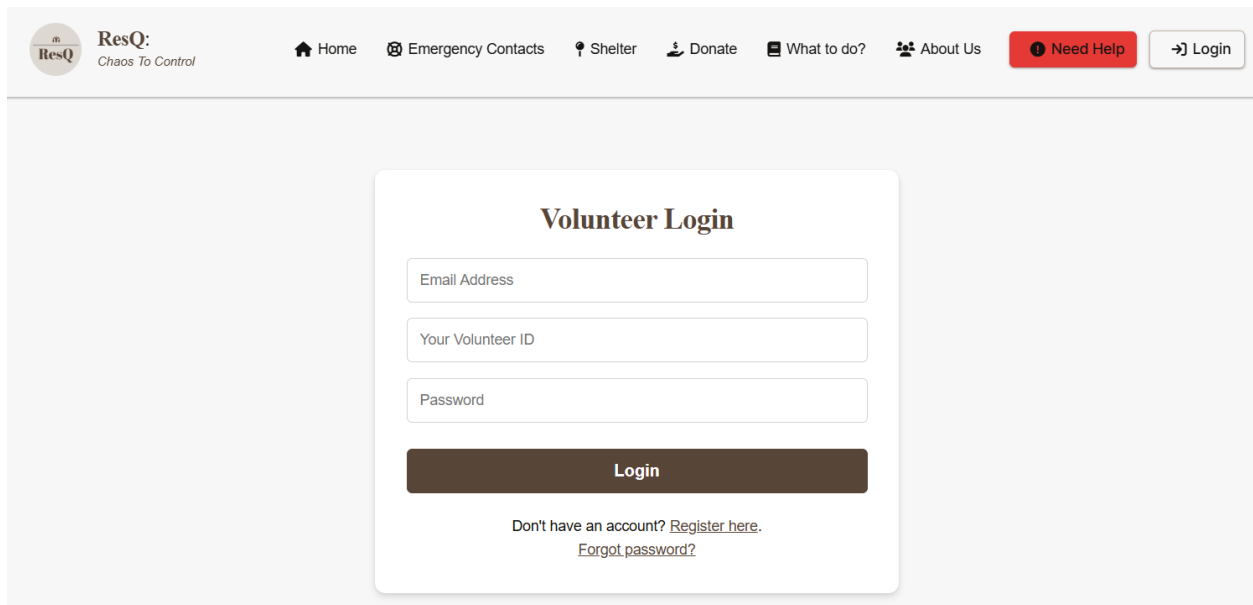


Fig 4.2. Role-based login

Volunteer Login:



The Volunteer Login form is centered on the page. It features a white background with a dark brown border. The title "Volunteer Login" is at the top in a dark brown font. Below the title are three input fields: "Email Address", "Your Volunteer ID", and "Password". A dark brown "Login" button is positioned below the input fields. At the bottom of the form, there is a link to "Register here" and a link to "Forgot password?". The form is set against a light gray background.

ResQ: Chaos To Control

Home Emergency Contacts Shelter Donate What to do? About Us Need Help Login

Volunteer Login

Email Address

Your Volunteer ID

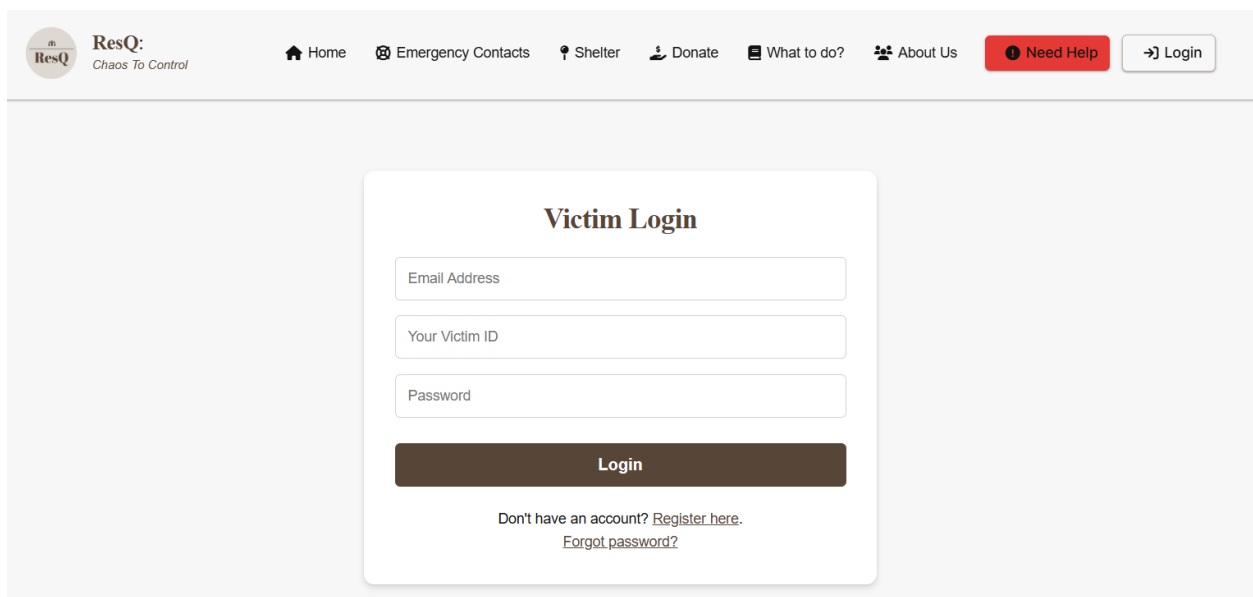
Password

Login

Don't have an account? [Register here.](#)
[Forgot password?](#)

Fig 4.3. Volunteer login

Victim Login



The Victim Login form is centered on the page. It features a white background with a dark brown border. The title "Victim Login" is at the top in a dark brown font. Below the title are three input fields: "Email Address", "Your Victim ID", and "Password". A dark brown "Login" button is positioned below the input fields. At the bottom of the form, there is a link to "Register here" and a link to "Forgot password?". The form is set against a light gray background.

ResQ: Chaos To Control

Home Emergency Contacts Shelter Donate What to do? About Us Need Help Login

Victim Login

Email Address

Your Victim ID

Password

Login

Don't have an account? [Register here.](#)
[Forgot password?](#)

Fig 4.4. Victim Login

Volunteer registration:

The screenshot shows the 'ResQ: Chaos To Control' website with a navigation bar containing links for Home, Emergency Contacts, Shelter, Donate, What to do?, About Us, Need Help, and Login. The main content area features a 'Register as a Volunteer' form with the instruction 'Kindly fill in this form to register.' The form includes input fields for Full Name, Email, Contact Number (e.g. 01712345678), Date of Birth (mm/dd/yyyy), Full Address (e.g. 123 Main St, City), Assigned Location (District, City), Occupation, Medical Info (or 'None'), Blood Group (e.g. A+), Allergies (or 'None'), and Chronic Disease (or 'None').

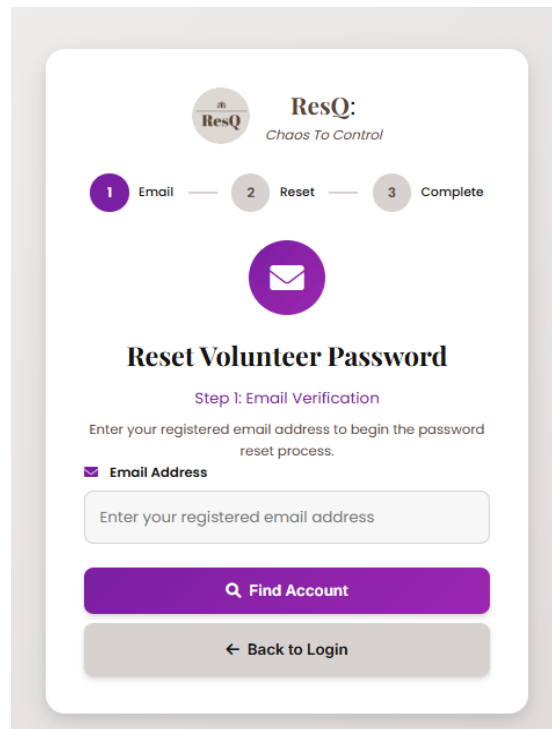
Fig 4.5. Volunteer registration

Victim registration:

The screenshot shows the 'ResQ: Chaos To Control' website with a navigation bar containing links for Home, About Us, Emergency Contacts, Shelter, Donate, What to do?, Need Help, and Login. The main content area features a 'Register as a Victim' form with the instruction 'Kindly fill in this form to register.' The form includes input fields for Full Name, Email, Contact Number (e.g. 01712345678), Date of Birth (mm/dd/yyyy), Full Address (e.g. 123 Main St, City), and Current Location (District, City).

Fig 4.6. Victim registration

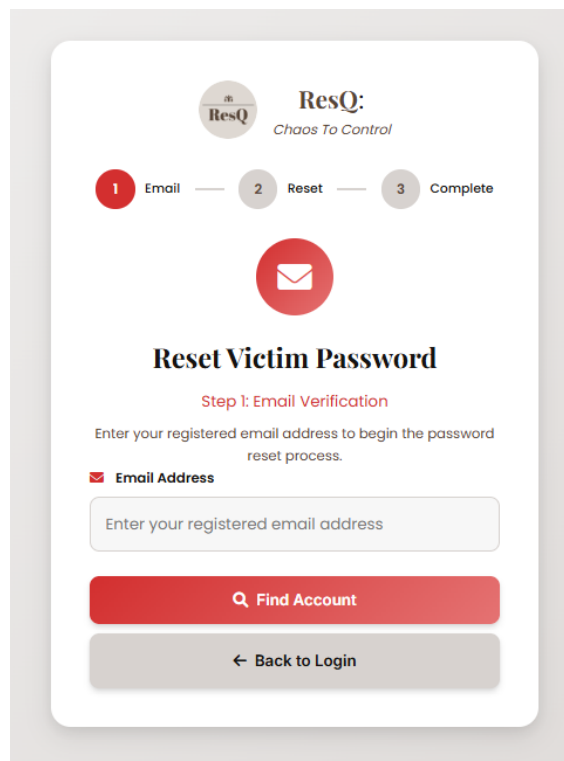
Volunteer Password Reset:



The form is titled "ResQ: Chaos To Control" with a logo. It features a progress bar with three steps: 1 Email (active), 2 Reset, and 3 Complete. Below the progress bar is a purple envelope icon. The main heading is "Reset Volunteer Password". Underneath, it says "Step 1: Email Verification" and "Enter your registered email address to begin the password reset process." There is a label "Email Address" with a checkmark icon. Below this is a text input field with the placeholder "Enter your registered email address". At the bottom, there are two buttons: a purple "Find Account" button with a magnifying glass icon, and a grey "Back to Login" button with a left arrow icon.

Fig 4.7. Volunteer Forget Password

Victim Password Reset:

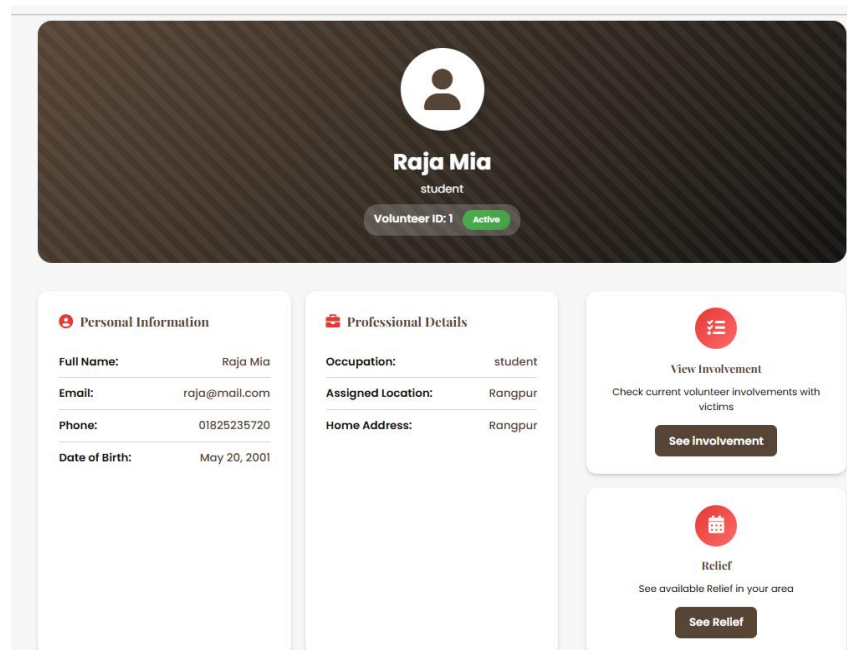


The form is titled "ResQ: Chaos To Control" with a logo. It features a progress bar with three steps: 1 Email (active), 2 Reset, and 3 Complete. Below the progress bar is a red envelope icon. The main heading is "Reset Victim Password". Underneath, it says "Step 1: Email Verification" and "Enter your registered email address to begin the password reset process." There is a label "Email Address" with a checkmark icon. Below this is a text input field with the placeholder "Enter your registered email address". At the bottom, there are two buttons: a red "Find Account" button with a magnifying glass icon, and a grey "Back to Login" button with a left arrow icon.

Fig 4.8. Victim Forget Password

4.2.3. Profile management

Volunteer Profile:



The Volunteer Profile for Raja Mia is displayed on a dark-themed header. The header includes a profile icon, the name "Raja Mia", the role "student", and a "Volunteer ID: 1" badge with an "Active" status. Below the header, the profile is divided into three columns. The first column, "Personal Information", lists: Full Name: Raja Mia, Email: raja@mail.com, Phone: 01825235720, and Date of Birth: May 20, 2001. The second column, "Professional Details", lists: Occupation: student, Assigned Location: Rangpur, and Home Address: Rangpur. The third column contains two interactive cards: "View Involvement" (Check current volunteer involvements with victims, See Involvement button) and "Relief" (See available Relief in your area, See Relief button).

Personal Information	
Full Name:	Raja Mia
Email:	raja@mail.com
Phone:	01825235720
Date of Birth:	May 20, 2001

Professional Details	
Occupation:	student
Assigned Location:	Rangpur
Home Address:	Rangpur

View Involvement

Check current volunteer involvements with victims

See Involvement

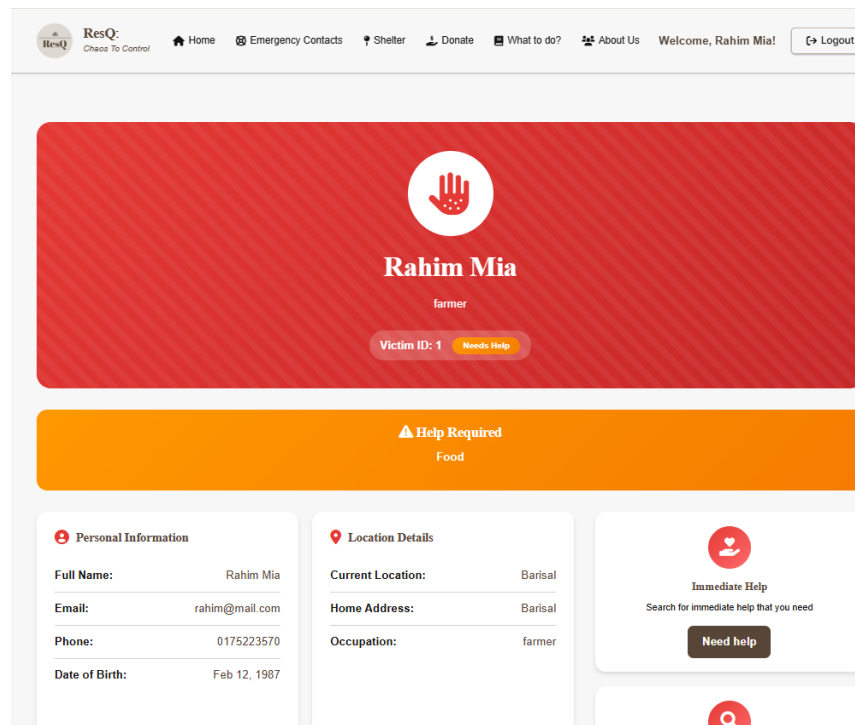
Relief

See available Relief in your area

See Relief

Fig 4.9. Volunteer Profile

Victim Profile:



The Victim Profile for Rahim Mia is displayed on a light-themed header. The header includes the ResQ logo, navigation links (Home, Emergency Contacts, Shelter, Donate, What to do?, About Us), a welcome message "Welcome, Rahim Mia!", and a "Logout" button. Below the header, the profile is divided into three columns. The first column, "Personal Information", lists: Full Name: Rahim Mia, Email: rahim@mail.com, Phone: 0175223570, and Date of Birth: Feb 12, 1987. The second column, "Location Details", lists: Current Location: Barisal, Home Address: Barisal, and Occupation: farmer. The third column contains two interactive cards: "Help Required" (Food) and "Immediate Help" (Search for immediate help that you need, Need help button). A search icon is visible at the bottom right.

Personal Information	
Full Name:	Rahim Mia
Email:	rahim@mail.com
Phone:	0175223570
Date of Birth:	Feb 12, 1987

Location Details	
Current Location:	Barisal
Home Address:	Barisal
Occupation:	farmer

Help Required

Food

Immediate Help

Search for immediate help that you need

Need help

Fig 4.10. Victim Profile

Volunteer Profile Update

The screenshot shows the 'Update Volunteer Profile' page. The header includes the ResQ logo, navigation links (Home, Donate, What to do?, Profile, Emergency), and a user profile section for 'Raja Mia' with a Logout button. The main heading is 'Update Volunteer Profile' with a sub-link 'Edit Your Information'. Below this is a form titled 'Edit Profile Information' with the instruction 'Update your volunteer details and preferences'. The form contains the following fields:

Field	Value
Full Name *	Raja Mia
Email Address *	raja@mail.com
Phone Number *	01825235720
Date of Birth	05/20/2001
Occupation	student
Blood Group	A+
Address	Rangpur

Fig 4.11. Volunteer Profile Update

Victim Profile Update

The screenshot shows the 'Update Victim Profile' page. The header is similar to the previous page but the user profile section is for 'Rahim Mia'. The main heading is 'Update Victim Profile' with a sub-link 'Edit Your Information'. Below this is a form titled 'Edit Profile Information' with the instruction 'Update your personal details and assistance needs'. The form contains the following fields:

Field	Value
Full Name *	Rahim Mia
Email Address *	rahim@mail.com
Phone Number *	0175223570
Date of Birth	02/12/1987
Occupation	farmer
Blood Group	B+
Address	Borisal
Current Location	
Assistance Requirements	

Fig. 4.12. Victim Profile Update

4.2.4. Emergency contact

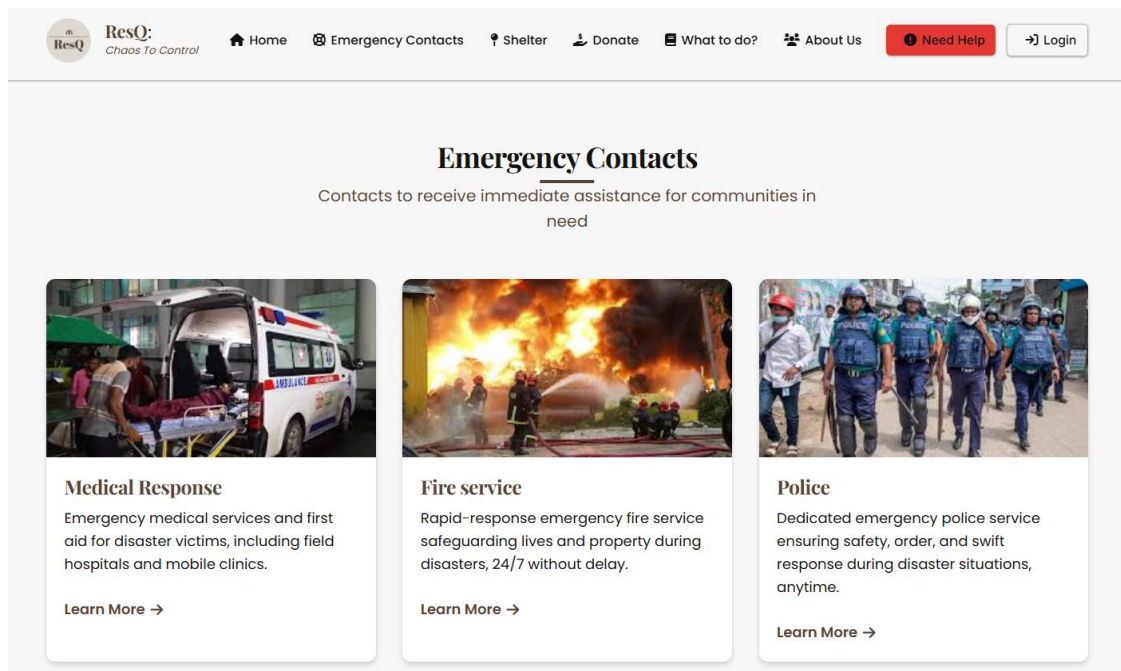


Fig. 4.13. Emergency Contact Dashboard

Medical Response:

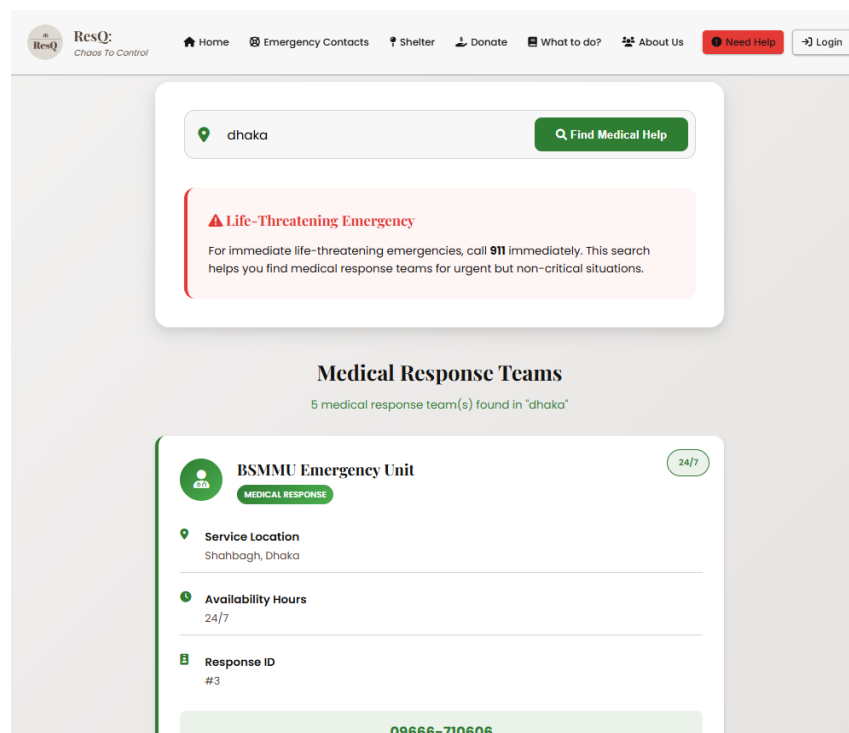


Fig. 4.14. Medic Response

Fire Service:

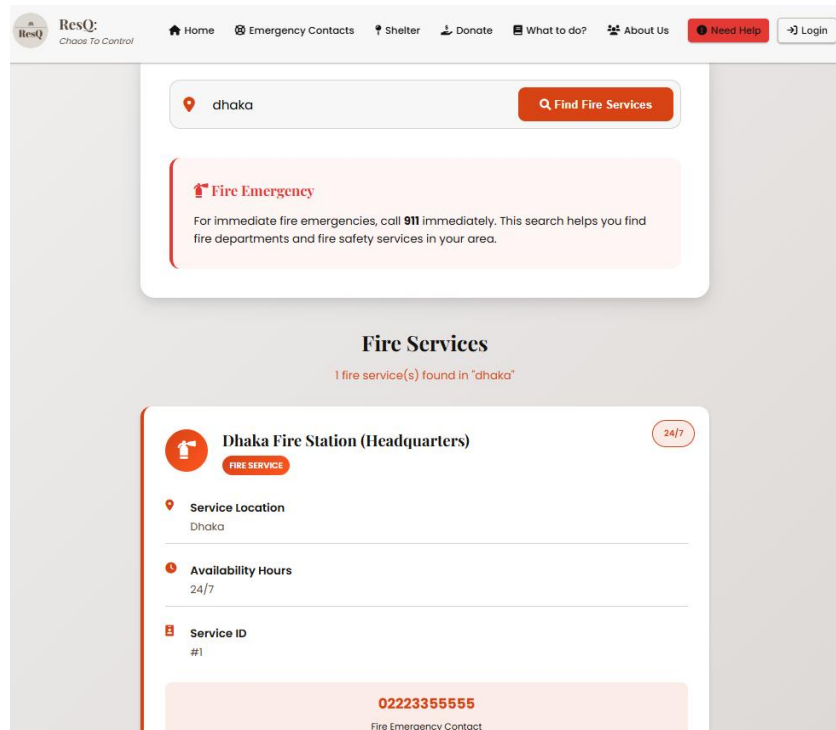


Fig. 4.15. Fire Service

Police:

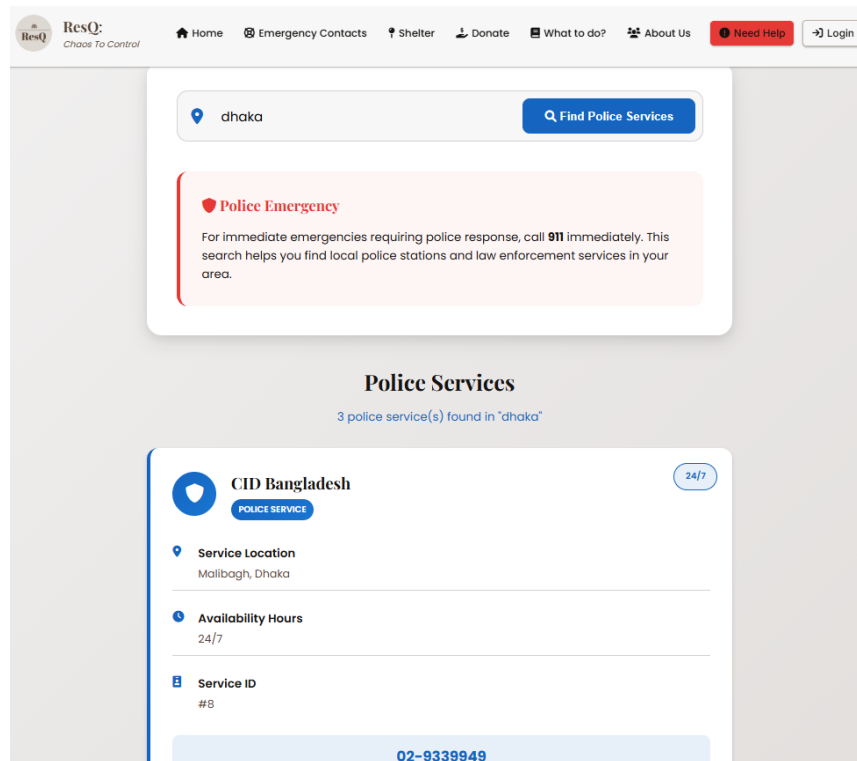


Fig. 4.16. Police

NGO Helpline:

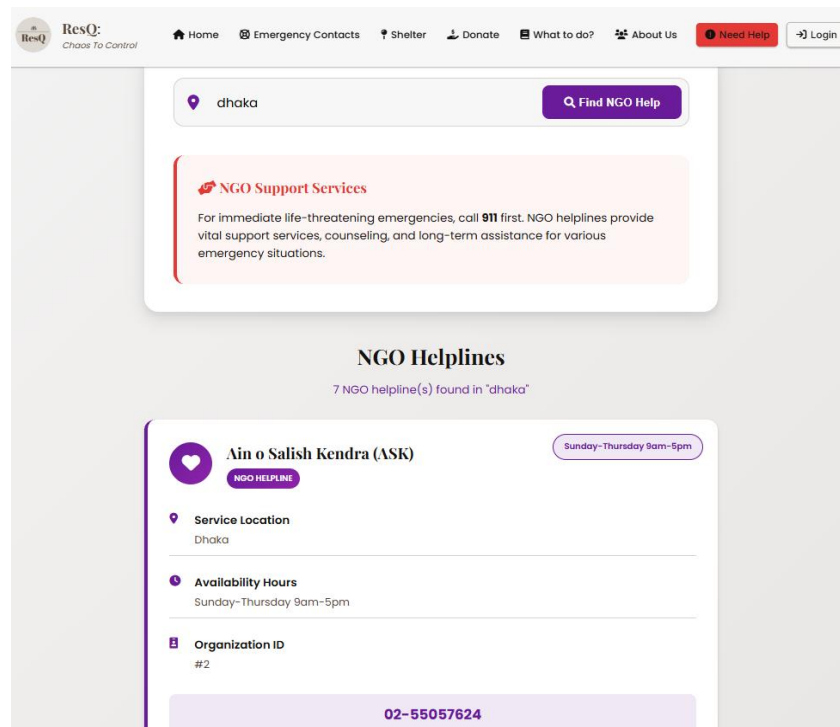


Fig. 4.17. NGO Helpline

Evacuation Support:

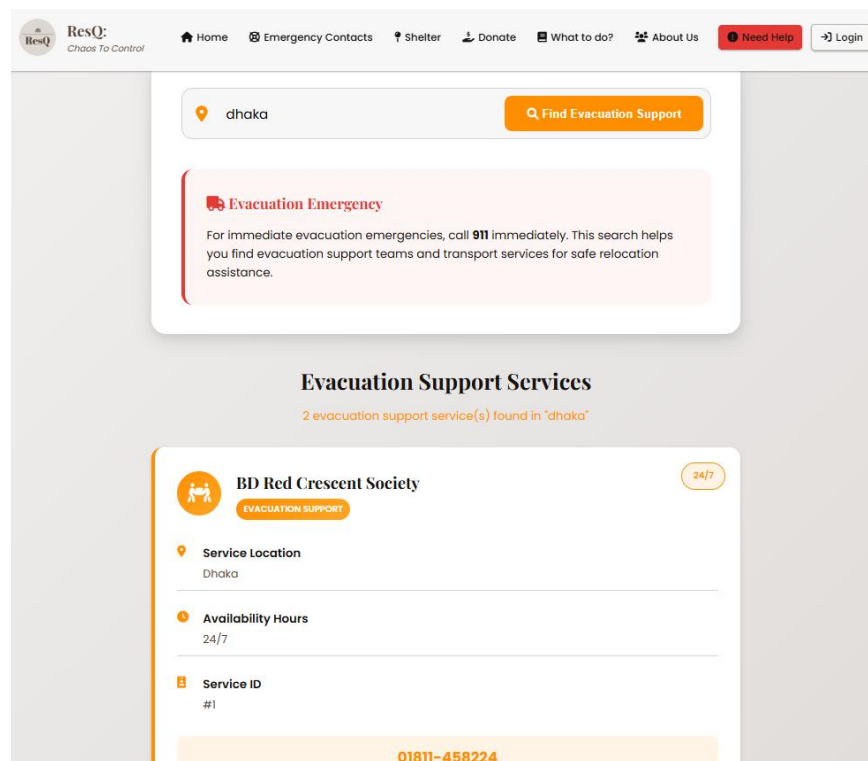


Fig. 4.18. Evacuation Support

Supply Distribution:

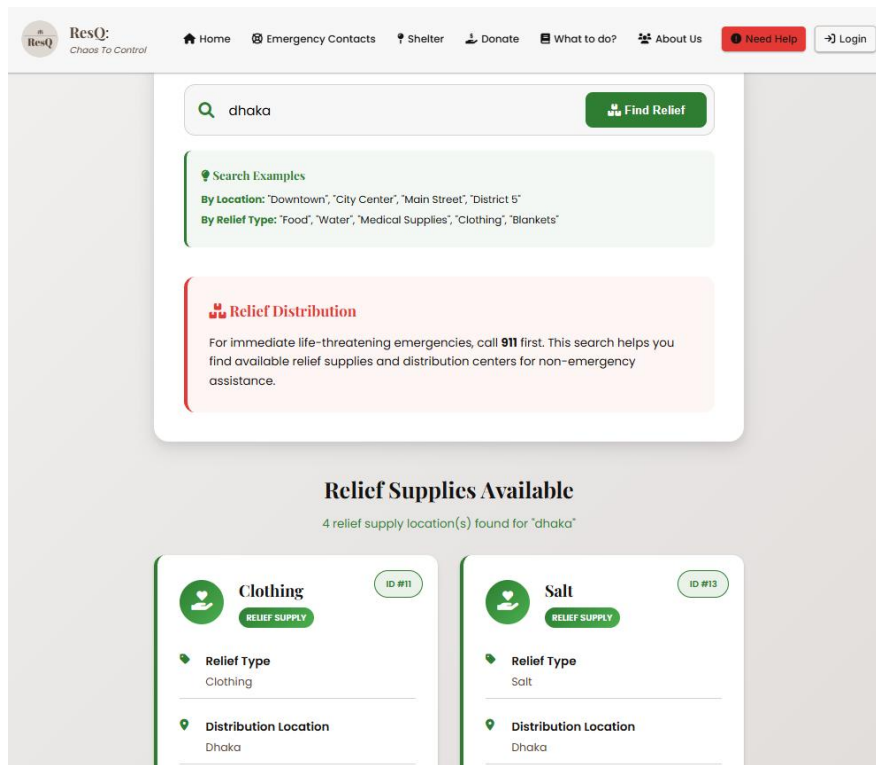


Fig. 4.19. Supply Distribution

4.2.5. Shelter Search

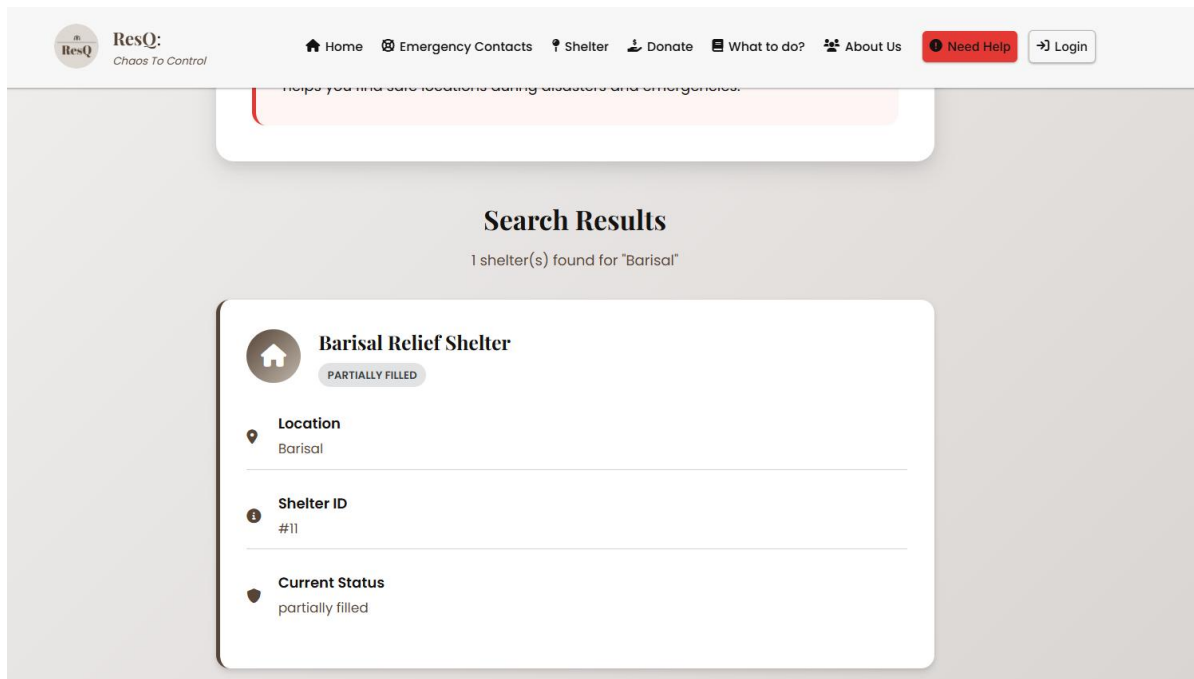


Fig. 4.20. Search Shelter

4.2.6. Disaster Response Guide (“What to do?”)

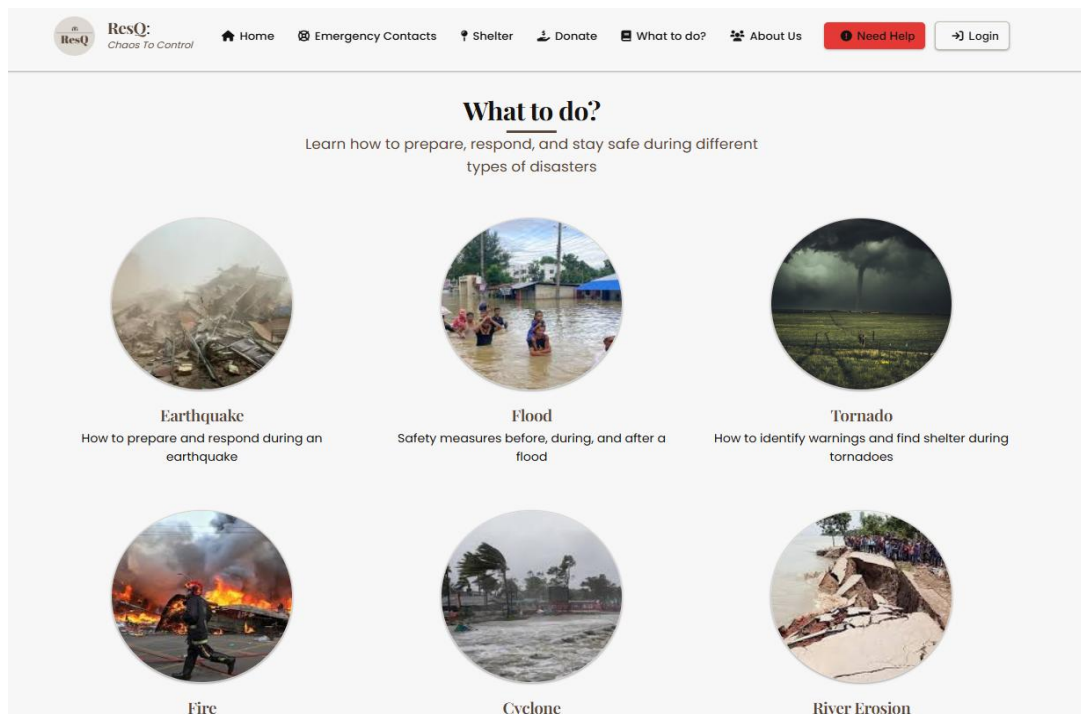


Fig. 4.21. “What to do?” Dashboard

Earthquake response guide:



Fig. 4.22. Earthquake guide

Flood response guide:

The screenshot shows the 'Flood Emergency Guide' page on the ResQ website. The header includes the ResQ logo and navigation links: Home, Emergency Contacts, Shelter, Donate, What to do?, About Us, Need Help, and Login. The main heading is 'Flood Emergency Guide' with the subtitle 'Quick actions to stay safe during floods in Bangladesh'. Below this are three buttons: '999 - Emergency', '1070 - Disaster Control', and '1090 - Flood Warning'. The 'Immediate Actions' section contains three cards: 'Move to Higher Ground' (with an upward arrow icon), 'Avoid Flood Waters' (with a car icon), and 'Turn Off Utilities' (with a plug icon). The 'Warning Signs & Preparation' section shows a list starting with '1 Heavy Rainfall'.

ResQ: Chaos To Control

Home Emergency Contacts Shelter Donate What to do? About Us Need Help Login

Flood Emergency Guide

Quick actions to stay safe during floods in Bangladesh

999 - Emergency 1070 - Disaster Control 1090 - Flood Warning

Immediate Actions

- Move to Higher Ground**
Get to the highest possible point immediately if flooding begins
- Avoid Flood Waters**
Never walk or drive through flood waters – just 15cm can sweep you away
- Turn Off Utilities**
Turn off electricity, gas and water at main switches/valves if safe to do so

Warning Signs & Preparation

- 1 Heavy Rainfall**

Fig. 4.23. Flood Guide

Tornado Response Guide:

The screenshot shows the 'Tornado Emergency Guide' page on the ResQ website. The header is identical to the flood guide. The main heading is 'Tornado Emergency Guide' with the subtitle 'Quick actions to stay safe during tornadoes in Bangladesh'. Below this are three buttons: '999 - Emergency', '1070 - Disaster Control', and '1096 - Weather Updates'. The 'Immediate Actions' section contains three cards: 'Seek Shelter Immediately' (with a house icon), 'Find Safe Spot' (with a book icon), and 'Protect Yourself' (with a person icon). The 'Warning Signs & Preparation' section shows a list starting with '1 Dark, Greenish Sky'.

ResQ: Chaos To Control

Home Emergency Contacts Shelter Donate What to do? About Us Need Help Login

Tornado Emergency Guide

Quick actions to stay safe during tornadoes in Bangladesh

999 - Emergency 1070 - Disaster Control 1096 - Weather Updates

Immediate Actions

- Seek Shelter Immediately**
Go to the lowest level of a sturdy building, preferably a basement or storm cellar
- Find Safe Spot**
Stay in an interior room or hallway away from windows, doors and outside walls
- Protect Yourself**
Cover yourself with blankets or get under sturdy furniture to protect from debris

Warning Signs & Preparation

- 1 Dark, Greenish Sky**
The sky may appear dark and greenish, which is a sign of severe weather

Fig.4.24. Tornado Guide

Fire Response Guide:

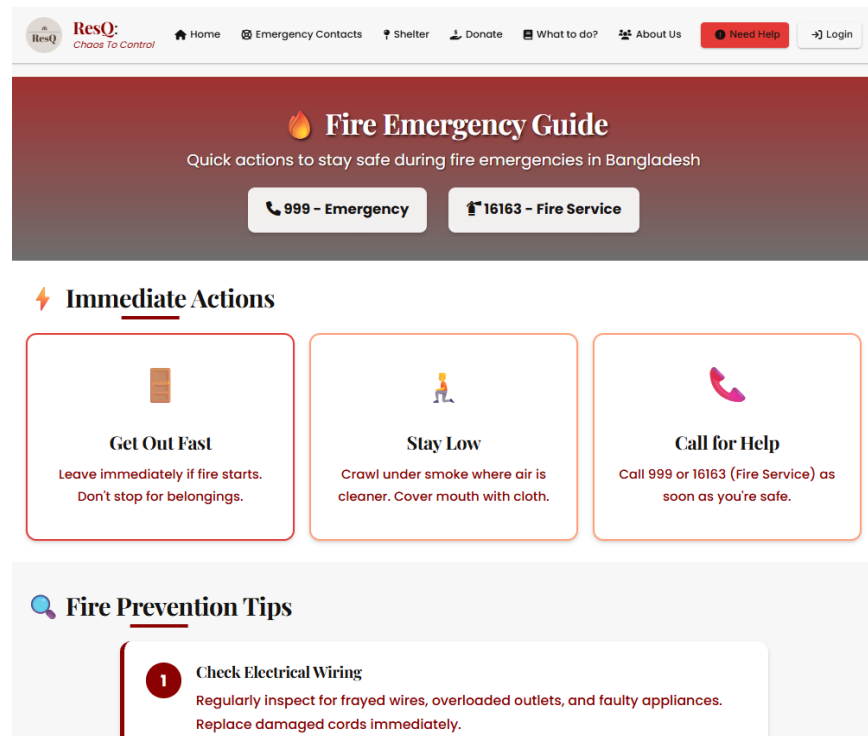


Fig. 4.25. Fire Response Guide

Cyclone Response Guide:

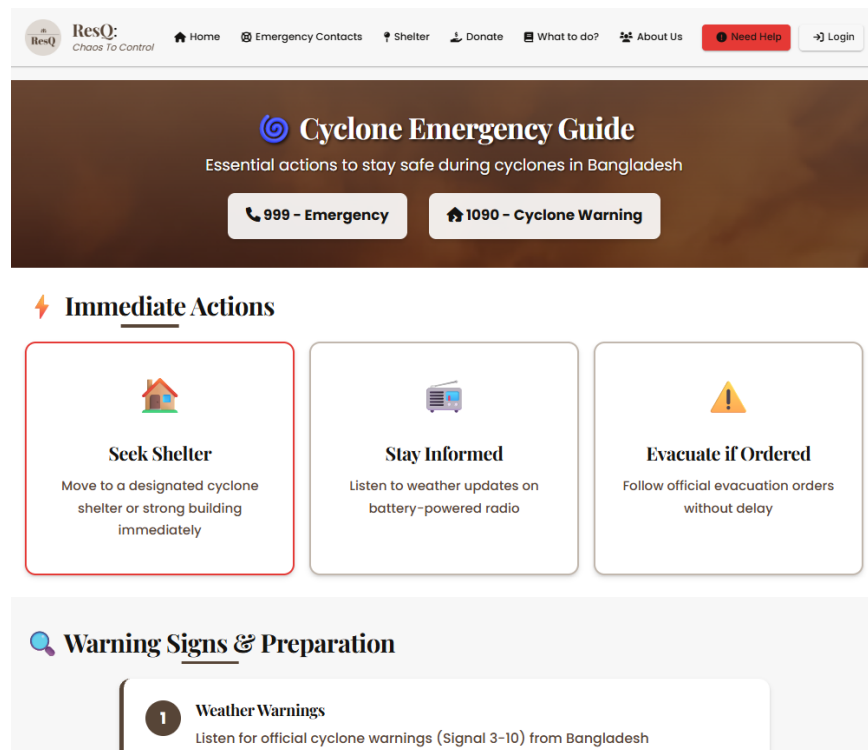


Fig. 4.26. Cyclone Guide

River Erosion Response Guide:

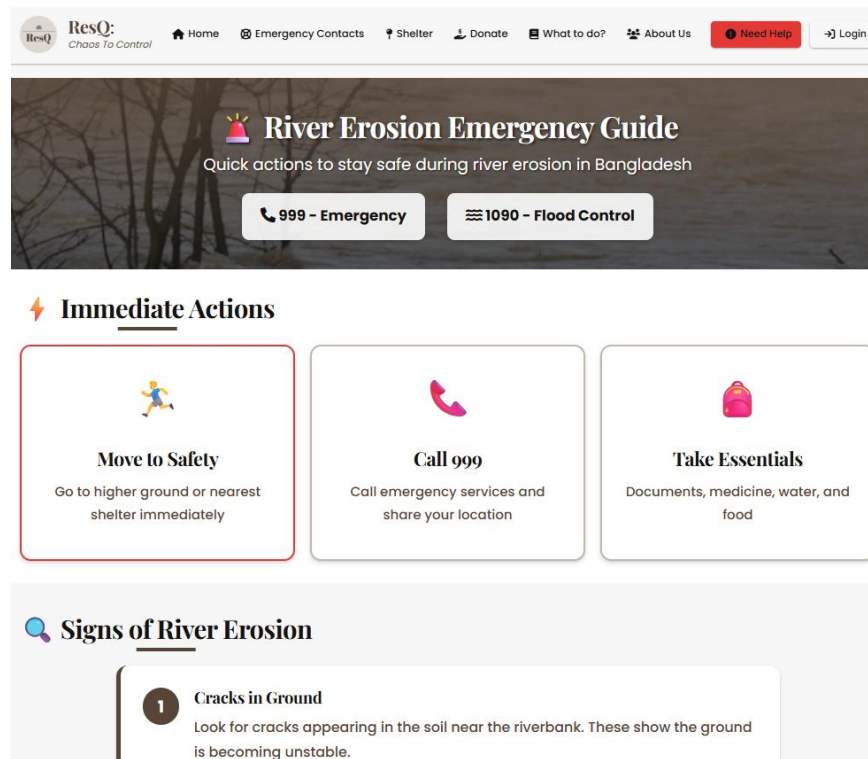


Fig. 4.27. River Erosion Guide

4.2.7. Immediate Helplines (“Need help”)

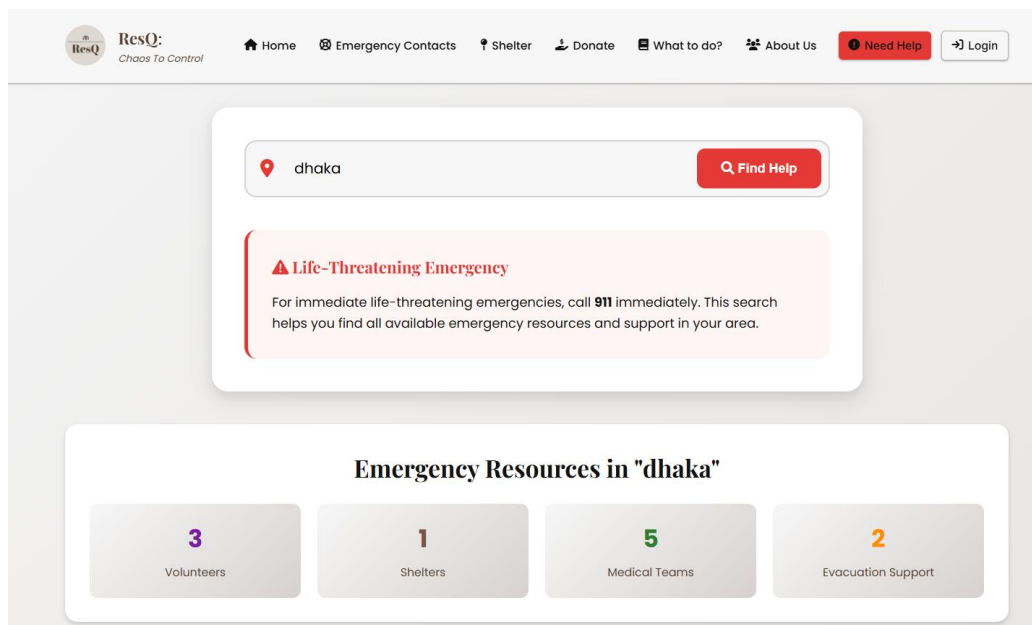


Fig. 4.28. “Need Help” Results

4.2.8. Donation Portal (Dummy for now)

Support Our Mission

Help us provide immediate disaster relief
Your donation helps us provide emergency response, supplies, and long-term recovery support to communities affected by disasters.

One-time Donation

৳500 ৳1,000 ৳2,000

৳5,000 ৳10,000 Custom

Custom Amount (৳)

Choose Payment Method

Credit/Debit Card bKash

Nogod Bank Transfer

Donate Now

Monthly Support

৳200 ৳500 ৳1,000

৳2,000 ৳5,000 Custom

Custom Monthly Amount (৳)

Choose Payment Method

Credit/Debit Card bKash

Nogod Bank Transfer

Start Monthly Support

Fig. 4.29. Donation Portal

4.2.9. View Involvement

VOLUNTEER	VICTIM	LOCATION	RELIEF DETAILS	STATUS
Nure Alam ID: 2 0178289525 nurealam@mail.com	Karim Mia ID: 2 0178226574	Dhaka	Rice ID: 1 Qty: 50 Kg Barisal	Active
Nure Alam ID: 2 0178289525 nurealam@mail.com	Karim Mia ID: 2 0178226574	Dhaka	Water ID: 2 Qty: 50 liter Dhaka	Active
Sadia ID: 9 0178256133 sadi@mail.com	Jala ID: 9 0175863189	Feni	Water ID: 2 Qty: 50 liter Dhaka	Active

Fig. 4.30. Involvement records

4.2.10. Search Volunteers

The screenshot shows the 'Search Volunteers' page on the ResQ: Chaos To Control website. The header includes the logo, navigation links (Home, Emergency Contacts, Shelter, Donate, What to do?, About Us), and buttons for 'Need Help' and 'Login'. The main heading is 'Search Volunteers' with the subtitle 'Find volunteers by name or location to coordinate disaster response'. A search bar contains 'Barisal' and a 'Search' button. Below the search bar, it states '2 volunteer(s) found matching "Barisal"'. Two volunteer profiles are displayed:

Volunteer Name	Location	Age	Occupation	Blood Group	Medical Info	Allergies	Call	Email
Lamia Nahar	Barisal Medical Area	30 years	Nurse	O+	None	None	Call	Email
Ahsan Ullah	Barisal	36 years	Farmer	B+	None	None	Call	Email

Fig. 4.31. Volunteer Search Results

4.2.11. Search Victims

The screenshot shows the 'Search Victims' page on the ResQ: Chaos To Control website. The header is identical to the previous page. The main heading is 'Search Victims' with the subtitle 'Find victims by name or location to coordinate disaster response'. A search bar contains 'Barisal' and a 'Search' button. Below the search bar, it states '5 victim(s) found matching "Barisal"'. Five victim profiles are displayed in a grid:

Victim Name	Location	Age	Occupation	Blood Group	Immediate Needs	Medical Info	Allergies	Call	Email	Help
Rahim Mia	Barisal	38 years	farmer	B+	Food		nut	Call	Email	Help
Hamza	Barisal	25 years	student	-				Call	Email	Help
Mahbub Rahman	Barisal	46 years	Fisherman	O+	Shelter	None	Dust	Call	Email	Help
Imran Hossain	Barisal	38 years	Fisherman							
Laila	Barisal	28 years	House wife							

Fig. 4.32. Victim Search Results

4.2.12. About Us Page

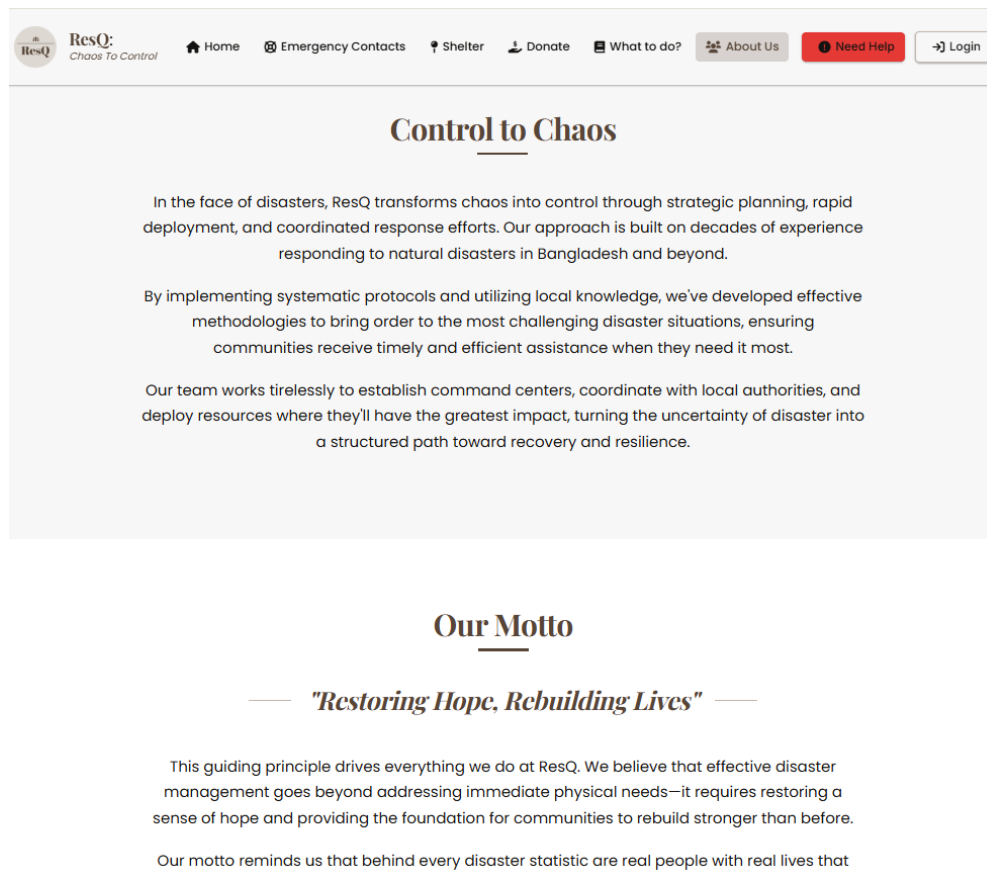


Fig. 4.33. “About Us” Page

4.3. Analysis and Interpretation

The ResQ: Chaos to Control web application was developed with the aim of providing an easy and efficient disaster management system. The research on the implemented system is based on its efficacy in meeting the set goals and providing the desired functionalities.

The essential modules of the system—assignment of users, resource assignment, finding shelter, coordination of volunteers, and relief assistance—were tested extensively in a local development environment using sample data. The result is that the system efficiently coordinates many pieces to enhance disaster response operations and facilitate stakeholder coordination.

4.3.1. System Functionality Testing

User Management:

The system supports role-based access for visitors, administrators, volunteers, and victims. Victims and volunteers can register and log in and be able to access respective functionalities based on their roles. Role-based redirection has been tested to ensure that users only access the modules they are allowed to access.

Victim-Volunteer Mapping:

Administrators can map volunteers to assist victims based on location and requirement easily. Dummy disaster data was considered to check this functionality, and the system could map volunteers to victims effectively.

Resource Allocation:

The inventory management system allows administrators to update and allocate relief resources efficiently. Tests were done to ensure that the system could handle updates in resource quantities and display the correct status to the users.

Shelter Search and Emergency Assistance:

The shelter finder feature provides victims with real-time information about the availability of shelters, locations. Similarly, emergency contact and rapid helpline modules enable users to get location-based contact information for critical services to enhance response in disaster cases.

Disaster Response Guidelines:

The system also offers common advice on disaster preparation accessible to all users, even site visitors. This is the function of making important safety information widely available to even non-registered persons on the system.

4.3.2. System Strengths

The system developed has a number of significant strengths that indicate its potential to become an effective disaster management system:

Easy-to-use Interface:

The user interface is sensitive and intuitive, hence simple for users with varying technical proficiency to navigate through. Bootstrap ensures perfect responsiveness on desktop computers,

tablets, and cell phones. Responsiveness is extremely essential in moments of catastrophe when individuals utilize other gadgets to obtain information.

Modular Architecture:

The modular structure of the system makes it easy to scale and maintain. Each module, such as user administration, resource allocation, and shelter location search, is created separately, meaning future developers can change or add specific functionalities without modifying the entire system.

Dynamic Data Handling:

With AJAX-based interactions built in, it minimizes the reloads of the page, making it more effective and faster to use. For example, bringing up emergency contacts or shelter capacity is instant without disrupting the workflow of the user.

Effective Data Management:

The PHP and MySQL-based server ensures safe storage and processing of the data. Validations and data sanitization will reduce errors and potential security vulnerabilities, especially when handling sensitive data about victims and volunteers.

Location-Specific Services

Features such as shelter search and emergency contacts provide location-based data based on the input user location, which is quite valuable in the coordination of on-site disaster relief operations.

4.3.3. System Evaluation

The system analysis indicates that ResQ successfully meets its primary objectives in the current prototype stage. The integration of primary modules creates a seamless coordination between victims, volunteers, and administrators.

Mock data testing made sure that workflows—from user registration to resource allocation—work properly. The ability of the system to fetch and display location-based shelters and emergency numbers confirms its functionality for actual usage.

Although the system is yet to be sold commercially, it shows great scalability potential. With future enhancements such as real-time GPS capability, auto-alerts, and cloud-based hosting with security, ResQ can also be a production-ready disaster management system capable of large-scale operations.

The findings report the success of the project in converting conceptual designs into a functional and cohesive web application, fulfilling the core requirements of disaster relief and response.

5. Discussion

The 'ResQ' project was designed particularly with one primary objective of establishing an orderly and complete solution towards efficient disaster handling. Its attention was mainly focused towards optimizing the operational efficiency of response towards disasters, as well as boosting transparency in regards to such vital efforts. By establishing a central platform, the 'ResQ' project offers an assortment of functionalities designed with specific intentions of easing and optimizing processes involved in distributing aid, coordinating volunteers, as well as accessibility towards immediate assistance in cases of disaster impact. During its course of execution, the project realized several notable achievements towards contributing to improved response in disasters. Nonetheless, it was subject to some of its drawbacks inherent in the project's perspective, as well as relying on technical means towards optimizing its functionality.

5.1. Outcomes of the project

The 'ResQ' project has accomplished with relative ease the production of several of its major results, whose application in everyday life, in everyday settings, can significantly improve as well as enhance in unprecedented ways the state of disaster management:

Streamlined Emergency Response:

The platform offers a notable increase in the deployment of aid and resources in a quick time frame through judiciously matching victims who require aid with volunteers who can assist them. By availing real-time connectivity of essential data, this platform can channel immediate assistance towards those regions in immediate need, without any sort of delay. This feature ensures the necessary resources arrive among affected populations in a timely as well as efficient way, thus maximizing the aggregated effect of relief efforts.

Enhanced Transparency in Relief Distribution:

Another one of 'ResQ's' greatest strengths is its unprecedented capacity to offer complete transparency in the complex process of relief distribution. By scrupulously following volunteer work as well as scrutinizing donor contributions, the system ensures that assistance is distributed in a justifiable manner and donors can witness just how their generous contributions are being utilized in real time. This transparency is responsible for generating immense levels of confidence

across all stakeholders involved, including not just donors per se but volunteers who work nonstop as well as disaster aid agencies working so hard to distribute assistance.

Coordination Among Stakeholders:

The project enhances and significantly raises the coordination efforts between diverse groups of individuals, such as victims, shelters, volunteers, as well as diverse agencies conducting relief works. This project is a crucial web-based portal that has been designed with care, with communication as its core focus with the effect of helping prevent any sort of confusion resulting in such scenarios and furthermore making it easy to engage in processes of emergency processes. Efficient and improved coordination is essential in highly decreasing response times, as well as assuring no geographic location or community is missed in such vital relief work being carried out.

Data-Driven Decision Making:

Despite it being worth mentioning that this feature is yet to be fully implemented in this latest release, its possible use in performing data analysis is a most significant result with immense potential towards future advancements. The portal has an inbuilt capability of collecting as well as analyzing diverse data of any kind, such as can subsequently go towards generating actionable intelligence vital towards efficient decision-making processes. By utilizing this data, decision-makers would ideally be able to prevent redundancy in operations, maximize allocation of resources in diverse regions, as well as pinpoint exact regions in need of aid most urgently, thus facilitating a fairer as well as just allocation of resources towards those most in need.

Public Empowerment:

'ResQ' is instrumental in enabling citizens to a great extent by giving them an umbrella facility under which they can report incidents of emergency, summon required assistance, as well as obtain crucial information on disasters. During crises, this facility is vital in helping people stay thoroughly informed as well as reach out to the larger population surrounding them. Connectivity is of maximum significance since it not only minimizes panic among citizens but is highly effective in deeming bases of misinformation obsolete with the setting up of a central feed establishing channels of credible news.

5.2. Limitations of the project

Although the 'ResQ' system does have a healthy amount of potential with it and offers a variety of opportunities towards further development, it is worth noting that there exist a number of flaws inherent in this specific variant. These flaws need detailed study as well as considerable corrective action before this system is successfully as well as usable in practical applications:

Requires Stable Internet Connectivity to Function:

The general working of the system depends extremely on the presence of a working and dependable internet connection. In emergency cases of severe disasters, in which the current infrastructure of any sort can be extremely broken or destroyed, it is not rare for reliable internet connectivity to be restricted or unavailable. In such specific cases, the system would have great difficulty working efficiently, thus posing problems both for victims in need of assistance as well as volunteers who are attempting to offer assistance, since both would have problems with reference to necessary information as well as in requesting relevant assistance.

Possible Delays in Data Entry During Emergencies:

Manual entry of a vast range of diverse data is a process encompassing essential details such as volunteer availability, shelter-specific locations, as well as diverse needs of assistance reported by victims. In such manual entry of diverse data, there can be immense delay, especially in high-stress environments wherein every moment is crucial and time is a nonrenewable resource. These delays, as they happen, significantly destabilize efficiencies in real-time responses whose purpose is served with such a system, eventually compromising timely aid deployment at those decisive points when it is most required.

Inconsistent Data Due to Manual Input:

Since the system is specifically dependent on several individuals responsible for entering data manually, it is truly significant to point out that the degree of precision and uniformity in the data received can vary considerably. These shifts in the quality of received data can lead to discrepancies in regards to several vital parameters, such as volunteer accessibility levels, details of victims, as well as data related to relief effort dispensation. In dealing with inconsistent data, there is some measure of risk involved in its potential complexities, leading to obstruction as well as delay of the response process as a whole.

Challenges in Remote or Network-Congested Areas:

For regions that have been most disrupted by disasters, it is not atypical for network congestion as well as under coverage to present huge problems, resulting in the system potentially facing a lot of hardship in working efficiently as well as effectively. With no reliable as well as reliable connectivity in such remote places, it can significantly hamper individuals' ability to effectively access the platform, thus hampering both communication initiatives as well as general relief work that is at such a crisis point.

6. Conclusion

The idea of ResQ was born out of the heart-rending floods in Feni when we understood how an unfortunate coordination breakdown and slow information generation can worsen an already difficult scenario. That experience led us to question: what if all sits on one platform where people can receive help, track resources, and share information during a crisis?

This project helped us turn that concept into a working system. We created a centralized platform for disaster management that integrates important services—emergency contact access, real-time shelter data, live tracking, and customizable dashboards. We aimed to better organize emergency response, make it more accessible and transparent, and we believe we made significant progress towards it.

Through working on ResQ, we had technical and collaboration problems. From resolving frontend and backend sync issues to working through Git conflicts when collaborating with others, each problem sharpened our development and collaboration skills. Most importantly, it was kept in mind the fact that technology, if well-designed, has the ability to actually impact human lives, especially in times of crisis.

7. Recommendations

Even when the system is running smoothly, there are some ways through which ResQ can expand as well. First of all, based on our experience during the Feni flood, we believe that the system can be made available as an offline-enabled mobile application. The majority of the affected areas do not have connectivity at the time of a disaster, and therefore, offline access would be critical.

We also recommend incorporating AI-based prediction applications to predict disaster patterns and improve relief distribution. This will make the platform proactive instead of reactive.

To make the platform inclusive, subsequent rollouts should come with the support of multiple languages and accessibility features that will enable everyone from all walks of life to use it comfortably.

Apart from that, we would also want IoT sensors, i.e., water or weather level sensors, to feed real-time environmental inputs to the system. That way, responses can be made even sooner.

We also want to explore having blockchain technology utilized in recording the distribution of aid, where supplies and donations could be traced publicly and securely, reducing the chance of mismanagement.

Lastly, we believe ResQ should be pilot-tested and utilized along with local NGOs and disaster management agencies, particularly in flood-prone areas like Feni. They could utilize their knowledge and on-ground experience to further modify the platform and prepare it to extend aid to communities whenever required.